

Mathematics

Year 6

Student Manual



Mathematics

YEAR 6 · STUDENT BOOK

Ages 10 to 11 · Upper Primary · First edition 2026

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Editorial Board

The **Prime Books Subject Series** has been developed under the guidance of the **Pedagogical Academic Group** for each subject and coordinated by the **Pedagogical Team of Prime School**.

This publication reflects our shared commitment to academic excellence, educational quality, professional integrity, and the continuous advancement of teaching and learning. Through the collective expertise and dedication of our educators, we aim to provide meaningful and engaging learning resources that support students' academic growth.

We extend our sincere appreciation to the **Pedagogical Department** and the **Content Creation Team of Prime School** for their invaluable contributions, dedication, and collaborative efforts in making this publication possible.

Welcome to Year 6 mathematics

This Student Book is your guide for mathematics in Year 6 at Prime School. It follows the Prime School Year 6 mathematics scheme of work for the full year: numbers, shapes, calculation, data, fractions, probability, ratio and transformation.

Each **unit** opens with what you will learn. Inside a unit, each **topic** teaches one idea carefully: a short explanation, at least one worked example, a second way to check, then practice you write in your exercise book.

You will meet decimals to three places, negative numbers, area, averages and more. The ideas build on Year 5. When something is new, the book shows the steps slowly before asking you to try.

How the year is organised

- **Term 1** · Units 1 to 6 · number sense, shapes and the four operations
- **Term 2** · Units 7 to 11 · averages, measures, angles, arithmetic laws and data
- **Term 3** · Units 12 to 17 · fractions, further calculation, probability, ratio and transformation

What you need most days

A pencil, a ruler and a squared exercise book. For some topics you will also need a protractor, a pair of compasses or a calculator when your teacher says so.

Work carefully. Check your answers. Talk through hard questions with a partner before you write. Mathematics in this book is something you *do*, not only something you read.

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© HOW TO USE THIS BOOK

Every panel has a job

Nothing on these pages is decoration. Learn the furniture once, then every unit feels familiar.

PANEL	WHAT IT DOES
Open the workshop	The idea the topic turns on, in two sentences
In this topic you will	The skills this topic teaches
Warm up	Talking before writing
Do you remember?	A link to earlier work
Here is how it works	The teaching, with tables or diagrams
Watch me work one out	A full example with every step
Check it a different way	A second method
Your turn	Practice in your exercise book
Watch out!	The mistake most pupils make here
Number fact	A real, checked fact (after the unit starts)
One step further	Extra challenge if you finish early

What you need for Unit 1

Ruler, pencil and squared exercise book. Keep a clear column for working. Other tools (compasses, protractor, calculator) appear later in the year when a topic needs them.

1

Place value and decimals

From a million down to thousandths

Every digit has a job. The job is its column.

🕒 **BY THE END OF THIS UNIT YOU WILL**

- ✓ explain the value of each digit in numbers with up to 3 decimal places
- ✓ read, write and compare numbers to a million and down to thousandths
- ✓ multiply and divide whole numbers and decimals by 10, 100 and 1000
- ✓ compose, decompose and regroup numbers with up to 3 decimal places
- ✓ round numbers with 2 decimal places to the nearest whole number and tenth

Unit 1 • Numbers

Place value: every digit has a column



Picture 1.0 Unit 1 map: ones, tenths, hundredths and thousandths. Each box is a column job.

■ OPEN THE WORKSHOP

In Year 6 we read numbers by **columns**, not only by digits. Move one column left and a digit is worth ten times as much. Move one column right and it is worth ten times less.

✦ WARM UP

Write these in figures, then read them to a partner:

- three hundred and six thousand and forty
- one million and five
- twenty point zero seven

Which was hardest, and why?

🔄 DO YOU REMEMBER?

In Year 5 you used tenths and hundredths. This year we add thousandths (0.001). The rule stays the same: each column is ten times the column on its right.

UNIT 1 - TOPIC 1.1

Place value to three decimal places

Here is how it works

Place value means the value of a digit depends on which column it stands in. The digit 6 is worth 600 in 643, but only 6 tenths (0.6) in 4.6.

Step 1 • Build a place-value chart for a large whole number

We use thin spaces in long numbers (for example 406 538) so groups of three digits are easy to read.

Table A • Whole-number place value for 406 538

HUNDRED THOUSANDS	TEN THOUSANDS	THOUSANDS	HUNDREDS	TENS	ONES
100 000	10 000	1000	100	10	1
4	0	6	5	3	8

Read it: **four hundred and six thousand, five hundred and thirty-eight.**

The digit **6** is in the thousands column, so its place value is **6000**. The digit **0** is in the ten thousands column. It is still needed: without it, 406 538 would look like 46 538.

Step 2 • Extend the chart through the decimal point

After the ones column come **tenths** (0.1), **hundredths** (0.01) and **thousandths** (0.001).

Table B • Decimal place value for 58.307

TENS	ONES	.	TENTHS	HUNDREDTHS	THOUSANDTHS
10	1	.	0.1	0.01	0.001
5	8	.	3	0	7

Read it: **fifty-eight point three zero seven.**

DIGIT	COLUMN	PLACE VALUE	IN WORDS
5	tens	50	fifty
8	ones	8	eight
3	tenths	0.3	three tenths
0	hundredths	0	zero hundredths
7	thousandths	0.007	seven thousandths

So $58.307 = 50 + 8 + 0.3 + 0.007$.

Step 3 • Compare decimals by columns, not by length

To compare 4.6 and 4.58:

- 1 Line up the decimal points.
- 2 Write 4.6 as 4.60 so both numbers show hundredths.
- 3 Compare tenths: both have 6 tenths.
- 4 Compare hundredths: 0 hundredths is less than 8 hundredths.
- 5 So **4.58 is smaller** than 4.6, even though 4.58 has more digits after the point.

✦ NUMBER FACT

A €2 coin is **2.20 mm** thick and a 1 cent coin is **1.67 mm** thick (European Central Bank). Those thicknesses are fixed to hundredths of a millimetre so machines can measure coins reliably. That is place value on a metal disc.

! WATCH OUT!

Never say a digit is "worth 3" without naming its column. In 58.307 the 3 is worth **0.3** (three tenths), not 3.

■ WATCH ME WORK ONE OUT

What is the place value of the 9 in 170.094?

- 1 Set out a place-value chart: hundreds, tens, ones, tenths, hundredths, thousandths.
- 2 The digits are 1, 7, 0, 0, 9, 4.
- 3 The 9 sits in the **hundredths** column.
- 4 Place value of the 9 = **0.09** (nine hundredths).

Answer: 0.09

⇌ CHECK IT A DIFFERENT WAY

Expand: $170.094 = 100 + 70 + 0.09 + 0.004$. The only term with 9 is **0.09**.

■ WATCH ME WORK ONE OUT

Which is larger, 6.3 or 6.28?

- 1 Write 6.3 as 6.30.
- 2 Tenths: both 3.
- 3 Hundredths: 0 compared with 8.
- 4 6.30 has fewer hundredths, so **6.3 is larger** than 6.28.

Answer: 6.3 is larger

◆ OUT IN THE REAL WORLD

At the Cascais fish counter the scale shows 1.245 kg. The 2 is tenths of a kilogram (0.2 kg), the 4 is hundredths (0.04 kg) and the 5 is thousandths (0.005 kg). Each digit on the display is a column on the chart.



Picture 1.1 A kitchen-style scale display. Notice tenths, hundredths and thousandths after the point: that is why 1.245 kg is not the same as 1.25 kg.

• YOUR TURN

- 1 What is the place value of the digit 7 in each number?
 - a. 670 346
 - b. 70.246
 - c. 3.075
 - d. 0.007
- 2 Write these numbers in figures.
 - a. four hundred and three thousand and six
 - b. two million and fifty
 - c. seventeen point zero eight
 - d. nine hundredths
- 3 Write these numbers in words.
 - a. 345 678
 - b. 1 002 040
 - c. 6.305
 - d. 0.08
- 4 Put these in order, smallest first: 3.5, 3.05, 3.55, 3.15, 3.005.
- 5 Write a decimal number that sits **between** 4.2 and 4.3.
- 6 Which is larger? Explain using column names.
 - a. 0.7 or 0.68
 - b. 12.09 or 12.1
 - c. 0.099 or 0.1
- 7 Inês reads a laboratory scale that shows 0.406 kg. What is the place value of the 6?

- 8 Write the number that has 3 hundreds, 0 tens, 5 ones, 2 tenths, 0 hundredths and 9 thousandths.

→ ONE STEP FURTHER

How many different numbers can you make that use each of the digits 2, 0, 5, 8 exactly once, with the point fixed after the first digit (like $a.bcd$)? List them in order from smallest to largest.

◆ THINK IT THROUGH

Can a number have **more digits** after the point and still be **smaller** than a number with fewer digits after the point? Give two examples and explain.

◆ OUT IN THE REAL WORLD

Gonçalo times a 100 m sprint on the school track. The stopwatch reads 12.08 s. His friend Tomás finishes in 12.8 s. Who was faster, and by how much, if you compare tenths and hundredths carefully?



Picture 1.2 Stopwatch times: 12.08 s and 12.8 s are different because the last digits sit in different columns.

▲ CHALLENGE YOURSELF

Using the digits 9, 3, 6, 8, 1 and 0 once each, and placing the decimal point where you choose, make:

- a. the largest number less than 100
- b. the smallest number greater than 10
- c. a number as close as possible to 50

Explain your placements.

UNIT 1 - TOPIC 1.2

Multiplying and dividing by 10, 100 and 1000

■ OPEN THE WORKSHOP

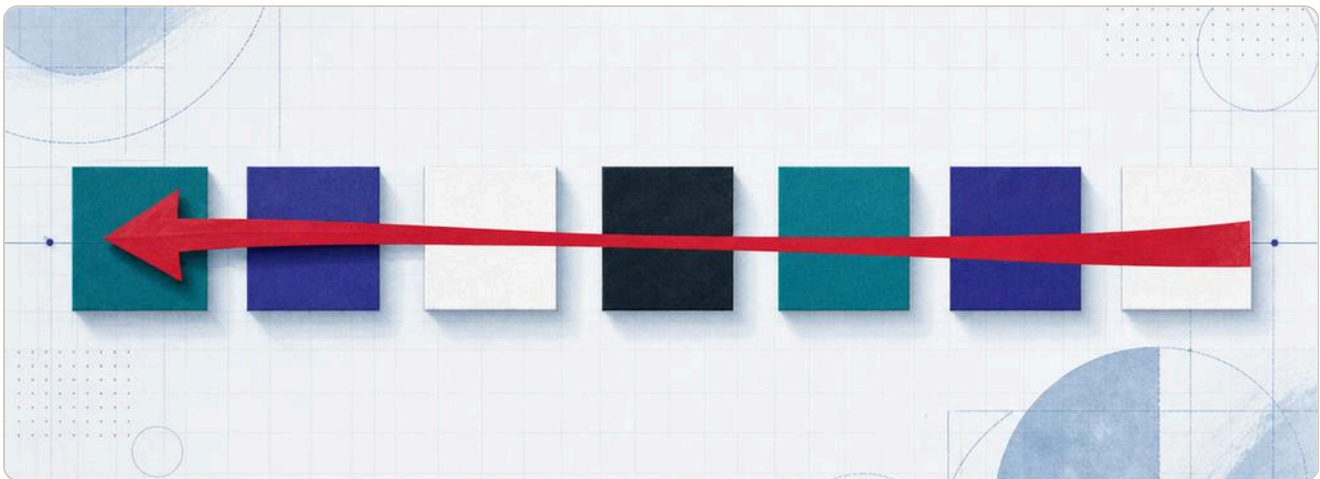
When you multiply by 10, every digit slides **one column to the left**. When you divide by 10, every digit slides **one column to the right**. The point stays still. The digits do the moving.

Here is how it works

OPERATION	HOW MANY COLUMNS THE DIGITS SLIDE	DIRECTION
× 10	1	left (bigger)
× 100	2	left
× 1000	3	left
÷ 10	1	right (smaller)
÷ 100	2	right
÷ 1000	3	right

Example: start with **3.407**.

CALCULATION	RESULT	WHAT MOVED
3.407×10	34.07	each digit one column left
3.407×100	340.7	each digit two columns left
3.407×1000	3407	each digit three columns left
$3.407 \div 10$	0.3407	each digit one column right
$3.407 \div 100$	0.03407	each digit two columns right



Picture 1.3 When you multiply by 100, each digit moves two columns left. The decimal point stays fixed.

! WATCH OUT!

Do **not** say "I moved the point". The point is the fixed hinge between ones and tenths. You moved the **digits**. Saying "I moved the point" works as a shortcut for whole numbers, and it fails as soon as you meet 0.034.

■ WATCH ME WORK ONE OUT

Work out 12.6×100 .

- 1** $\times 100$ means slide every digit **two** columns left.
- 2** 1 goes from tens to thousands; 2 goes from ones to hundreds; 6 goes from tenths to tens.
- 3** Fill empty columns with zeros if you need them for place value when writing whole numbers.

Answer: 1260

↔ CHECK IT A DIFFERENT WAY

$$12.6 \times 100 = 12.6 \times 10 \times 10.$$

$$12.6 \times 10 = 126$$

$$126 \times 10 = 1260$$

Same answer, in two single slides instead of one double slide.

■ WATCH ME WORK ONE OUT

Work out $7.05 \div 100$.

- 1** $\div 100$ means slide every digit **two** columns right.
- 2** 7 ones become 7 hundredths; 0 tenths become 0 thousandths; 5 hundredths become 5 ten-thousandths.
- 3** Write the result carefully: **0.0705**.

Answer: 0.0705

✦ NUMBER FACT

A car odometer that shows tenths of a kilometre is multiplying the number of full kilometres by 10 in the last drum. When the tenths drum finishes a full turn, the ones drum clicks forward by 1. That is place value in steel.



Picture 1.4 An odometer is a place-value machine made of drums.

• YOUR TURN

- 1 Multiply each number by 10, by 100 and by 1000.
 - a. 4.7
 - b. 0.36
 - c. 52.08
 - d. 0.009
- 2 Divide each number by 10, by 100 and by 1000.
 - a. 80
 - b. 6.3
 - c. 415
 - d. 9.01
- 3 Copy and complete.
 - a. $0.45 \times \underline{\quad} = 45$
 - b. $3.8 \times \underline{\quad} = 3800$
 - c. $720 \div \underline{\quad} = 7.2$
 - d. $9 \div \underline{\quad} = 0.009$
- 4 A laboratory pipette delivers 0.025 litres. How many millilitres is that?
(1 litre = 1000 millilitres.)
- 5 A school run is 2.5 km. How many metres is that?
- 6 True or false? Correct any false statement.
 - a. Multiplying by 100 always adds two zeros on the end.
 - b. $0.7 \times 10 = 7$
 - c. $4.05 \div 10 = 0.405$
 - d. $60 \div 1000 = 0.06$
- 7 Inês multiplies a number by 100 and gets 308. What was her starting number?

- 8 Write a number which, when divided by 1000, gives 0.012.

⇌ TALK ABOUT IT

Inês: "I moved the point two places."

Gonçalo: "The digits moved, not the point."

Who is describing the model this book uses, and why does it matter when the answer is less than one?

→ ONE STEP FURTHER

Find every way to fill the blanks with 10, 100 or 1000 so the statement is true:

$$0.6 \times ___ \div ___ = 6$$

▲ CHALLENGE YOURSELF

A number is multiplied by 10, then divided by 100, then multiplied by 1000. The final result is 45.6. What was the original number? Show the three steps in reverse.

UNIT 1 - TOPIC 1.3

Composing, decomposing and regrouping

■ OPEN THE WORKSHOP

Decompose means take a number apart into the value of each column.

Compose means put the parts back together. **Regroup** means write the same number using a different column unit, for example as tenths or as hundredths.

Here is how it works

Decompose 4.65:

$4.65 = 4 + 0.6 + 0.05$

Or, naming the columns:

$4 \text{ ones} + 6 \text{ tenths} + 5 \text{ hundredths}$

Compose from parts:

$200 + 40 + 0.3 + 0.07 = 240.37$

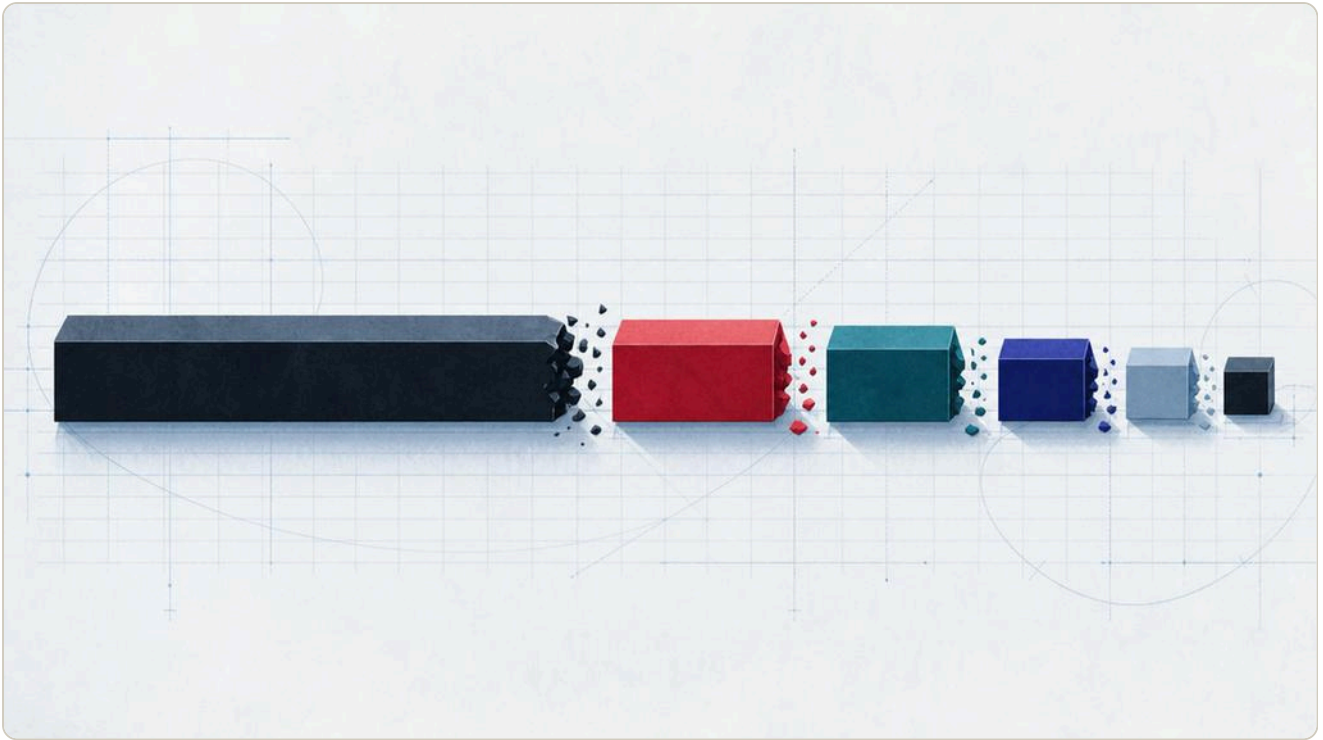
Regroup as tenths:

$4.65 = 46 \text{ tenths} + 5 \text{ hundredths}$, or simply **46.5 tenths**.

Regroup as hundredths:

$4.65 = 465 \text{ hundredths}$.

NUMBER	AS TENTHS	AS HUNDREDTHS	AS THOUSANDTHS
3.7	37 tenths	370 hundredths	3700 thousandths
0.62	6.2 tenths	62 hundredths	620 thousandths
1.5	15 tenths	150 hundredths	1500 thousandths



Picture 1.5 Composing and decomposing: the same value, different parts.

■ **WATCH ME WORK ONE OUT**

Regroup 0.308 as thousandths.

- 1 Thousandths are the third column after the point.
- 2 0.308 means 3 tenths + 0 hundredths + 8 thousandths.
- 3 3 tenths = 300 thousandths, so altogether **308 thousandths**.

Answer: 308 thousandths

⇒ **CHECK IT A DIFFERENT WAY**

Multiply by 1000 to count thousandths: $0.308 \times 1000 = 308$. That works because $\times 1000$ slides every digit three columns left, which is exactly "how many thousandths".

! WATCH OUT!

$0.4 = 4$ tenths, not 4 hundredths. If you write 0.4 as hundredths you need **40 hundredths**. Missing zeros are the most common regrouping error.

• **YOUR TURN**

- 1 Decompose each number into the value of every occupied column.
 - a. 4.65
 - b. 0.308
 - c. 27.4
 - d. 130.05
 - e. 9.999
 - f. 0.07
- 2 Compose each number from its parts.
 - a. $60 + 8 + 0.5 + 0.02$
 - b. $200 + 0.4 + 0.06$
 - c. $7 + 0.9 + 0.003$
 - d. $1000 + 30 + 0.7$
- 3 Regroup each number as tenths.
 - a. 4.6
 - b. 0.9
 - c. 12.3
 - d. 5
- 4 Regroup each number as hundredths.
 - a. 0.35
 - b. 1.06
 - c. 2.4
 - d. 0.8

- 5 Regroup each number as thousandths.
- a. 0.125
 - b. 0.4
 - c. 1.02
 - d. 3
- 6 Copy and complete.
- a. $3.7 = \underline{\hspace{1cm}}$ tenths
 - b. $0.62 = \underline{\hspace{1cm}}$ hundredths
 - c. $1.5 = \underline{\hspace{1cm}}$ thousandths
 - d. 40 hundredths = $\underline{\hspace{1cm}}$ (as a decimal)
- 7 Leonor has 250 hundredths. Write her amount as a decimal number of ones.
- 8 Which of these is equal to 0.7?
- a. 7 tenths
 - b. 70 hundredths
 - c. 700 thousandths
 - d. all of the above

→ **ONE STEP FURTHER**

Write 2.05 in **three** different ways: as a sum of place values, as hundredths, and as thousandths.

▲ **CHALLENGE YOURSELF**

Using only the parts 0.001, 0.01, 0.1 and 1, compose 3.147 in two different ways (different numbers of each part). Prove both ways total 3.147.

UNIT 1 - TOPIC 1.4

Rounding to the nearest whole number

OPEN THE WORKSHOP

Rounding to the nearest whole number asks one question: which whole number is this value **closer to**? Look at the **tenths** digit to decide.

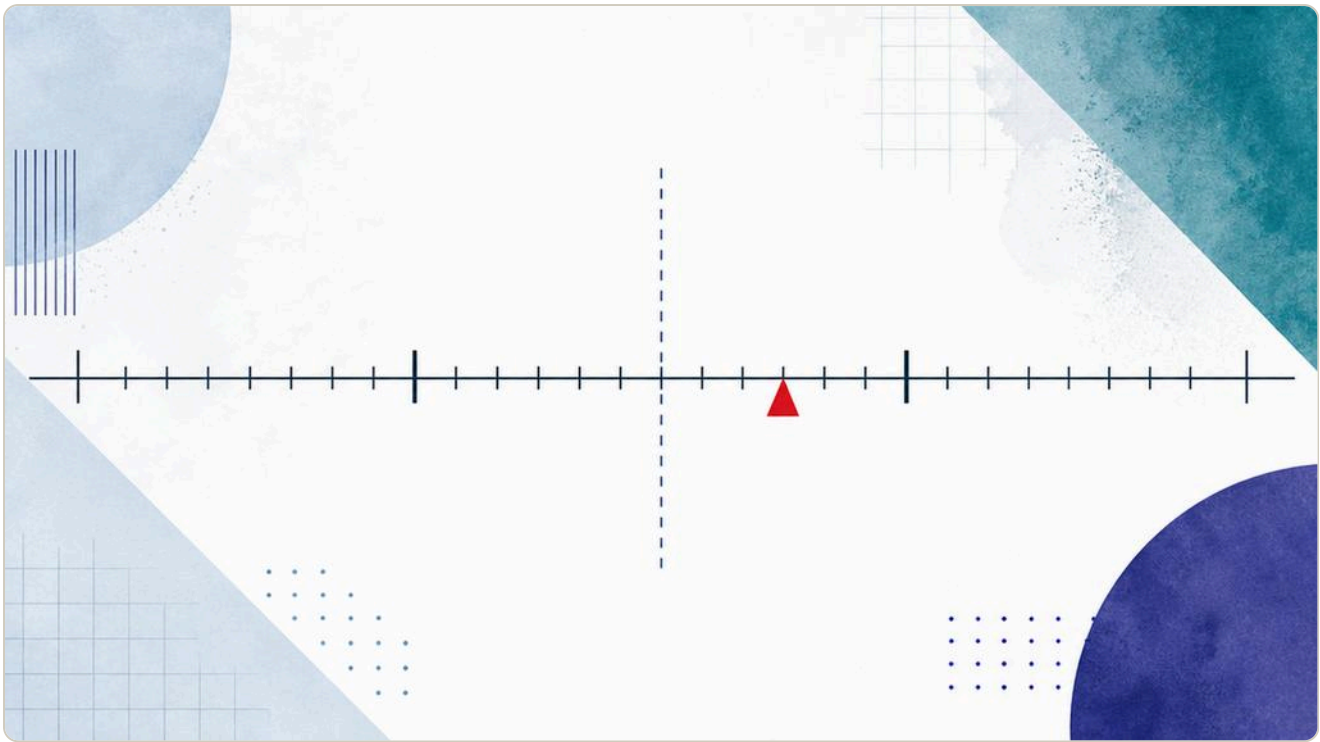
Here is how it works

To round a number with decimals to the nearest whole number:

- 1 Find the ones digit (the whole-number part).
- 2 Look at the **tenths** digit (the first digit after the point).
- 3 If the tenths digit is **0, 1, 2, 3 or 4**, round **down** (keep the ones).
- 4 If the tenths digit is **5, 6, 7, 8 or 9**, round **up** (ones go up by 1).

Halfway rule used in this book: **5 rounds up**. So 3.5 rounds to 4.

NUMBER	TENTHS DIGIT	NEAREST WHOLE
7.2	2	7
7.5	5	8
7.8	8	8
12.49	4	12
12.50	5	13
0.48	4	0
0.51	5	1



Picture 1.6 Rounding: decide which side of halfway the number sits on.

■ **WATCH ME WORK ONE OUT**

Round 34.62 to the nearest whole number.

- 1 Ones digit: 4 (the number is thirty-four and a bit).
- 2 Tenths digit: 6.
- 3 6 is 5 or more, so round up.

Answer: 35

⇒ **CHECK IT A DIFFERENT WAY**

Place 34.62 on a number line between 34 and 35. The halfway mark is 34.5. 34.62 is to the right of 34.5, so it is closer to 35. Same answer without the "rule song", because the line shows the distance.

! WATCH OUT!

When you round 9.7 to the nearest whole number, the ones digit is 9 and tenths is 7, so you round up to **10**, not to 0. Carrying into the next column is normal.

✦ NUMBER FACT

Digital kitchen scales often display mass to the nearest gram. If a scale that rounds to the nearest whole gram shows 3 g, the true mass could be anywhere from 2.5 g up to (but not including) 3.5 g. Rounding hides a small range.

• YOUR TURN

- 1 Round each number to the nearest whole number.
 - a. 6.2
 - b. 6.5
 - c. 6.8
 - d. 0.4
 - e. 0.5
 - f. 19.49
 - g. 19.5
 - h. 99.7
- 2 A packet of rice is labelled 1.25 kg. What is that mass to the nearest kilogram?
- 3 Match each number to its nearest whole: 2.4, 2.5, 3.49, 3.5, 3.51.
- 4 True or false?
 - a. 8.5 rounds to 8
 - b. 0.49 rounds to 0
 - c. 14.99 rounds to 15
 - d. 5.05 rounds to 5
- 5 Write two different numbers with two decimal places that both round to 12.
- 6 Write a number with two decimal places that is greater than 7.5 but still rounds to 7. Is that possible? Explain.
- 7 Duarte runs 3.48 km. His coach records the distance to the nearest kilometre. What does the coach write?
- 8 A shop price of €9.50 is rounded to the nearest euro for a charity tally. How many euros are tallied for one item?

→ ONE STEP FURTHER

List every number with **one** decimal place that rounds to 10. How many are there?

▲ CHALLENGE YOURSELF

Find a number with two decimal places that rounds to 5 as a whole number, but would round to 4.9 if you first rounded it to the nearest tenth. Is that possible? Prove it or show why not.

UNIT 1 - TOPIC 1.5**Rounding to the nearest tenth****■ OPEN THE WORKSHOP**

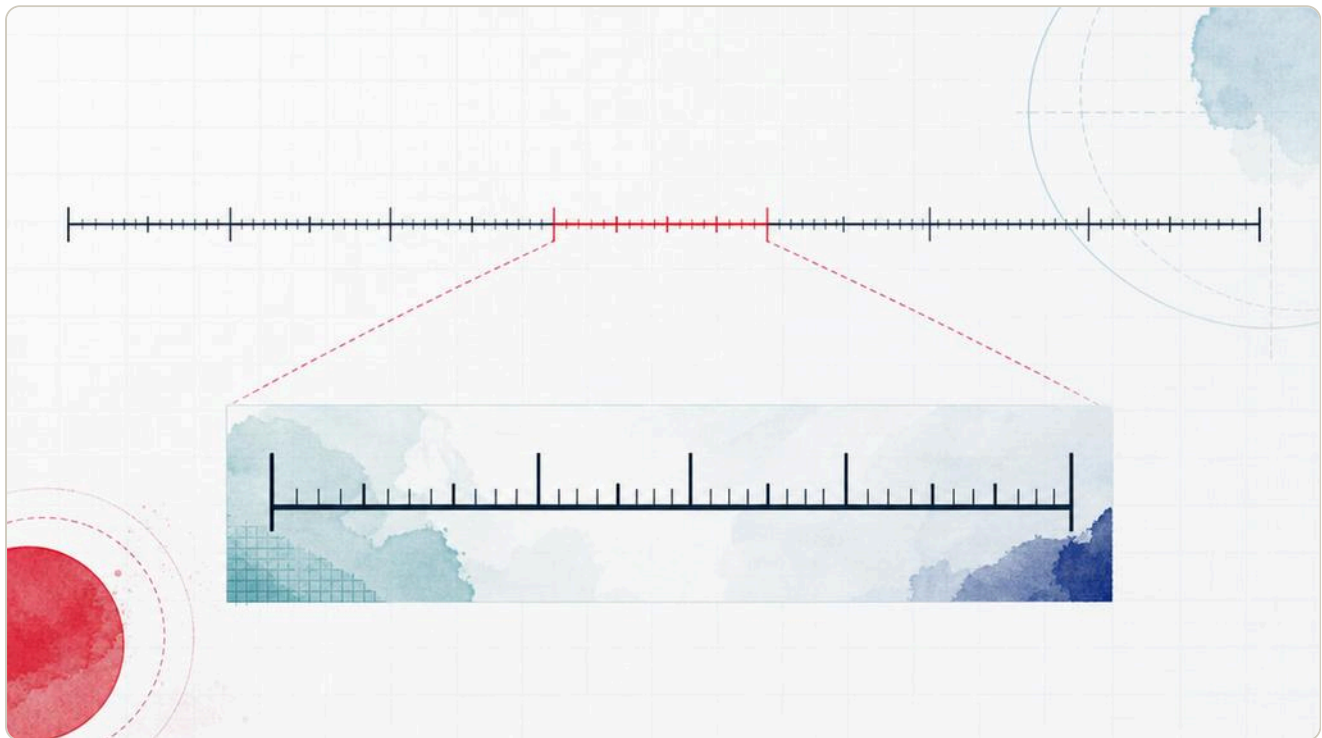
Rounding to the nearest tenth is the same decision as rounding to a whole number, but **ten times more precise**. Look at the **hundredths** digit to decide.

Here is how it works

To round to the nearest tenth:

- 1 Find the tenths digit.
- 2 Look at the **hundredths** digit (second digit after the point).
- 3 If hundredths are **0 to 4**, keep the tenths digit.
- 4 If hundredths are **5 to 9**, increase the tenths digit by 1.
- 5 Drop the digits after the tenths (or write them as 0).

NUMBER	HUNDREDTHS	NEAREST TENTH
3.42	2	3.4
3.45	5	3.5
3.48	8	3.5
12.04	4	12.0
12.05	5	12.1
0.96	6	1.0
7.95	5	8.0



Picture 1.7 To the nearest tenth: the same halfway rule, one column further in.

■ WATCH ME WORK ONE OUT

Round 6.35 to the nearest tenth.

- 1 Tenths digit: 3.
- 2 Hundredths digit: 5.
- 3 5 rounds up, so tenths become 4.

Answer: 6.4

⇌ CHECK IT A DIFFERENT WAY

On a number line from 6.3 to 6.4, halfway is 6.35. This book rounds halfway **up**, so 6.35 goes to 6.4. (Some scientific contexts use other halfway conventions; in this student book, 5 always rounds away from the smaller option.)

■ WATCH ME WORK ONE OUT

Round 9.96 to the nearest tenth.

- 1 Tenths digit: 9.
- 2 Hundredths digit: 6, so round up.
- 3 Tenths $9 + 1$ means the ones increase: **10.0**.

Answer: 10.0

! WATCH OUT!

Rounding 2.95 to the nearest tenth is **3.0**, not 2.10. When tenths roll past 9, you carry into the ones.

• YOUR TURN

- 1 Round each number to the nearest tenth.
 - a. 4.12
 - b. 4.15
 - c. 4.19
 - d. 0.74
 - e. 0.75
 - f. 18.04
 - g. 18.05
 - h. 9.99
- 2 Round 12.08 seconds to the nearest tenth of a second.
- 3 A fish counter shows 1.245 kg. What is that mass to the nearest tenth of a kilogram?
- 4 True or false?
 - a. 3.14 rounded to the nearest tenth is 3.1
 - b. 3.15 rounded to the nearest tenth is 3.1
 - c. 0.05 rounded to the nearest tenth is 0.1
 - d. 7.95 rounded to the nearest tenth is 8.0
- 5 Write two numbers with two decimal places that both round to 5.6 as the nearest tenth.
- 6 Which whole numbers can 4.5 round to? Which tenths can 4.55 round to?
- 7 Matilde measures a ribbon as 0.84 m. Write the length to the nearest tenth of a metre, and to the nearest whole metre.
- 8 Order these by size after each is rounded to the nearest tenth:
2.44, 2.45, 2.46, 2.41, 2.49.

→ ONE STEP FURTHER

Explain, with a number line sketch in your book, why 0.05 and 0.04 round to different tenths.

▲ CHALLENGE YOURSELF

A number has two decimal places. Rounded to the nearest tenth it is 7.2. Rounded to the nearest whole number it is 7. List all possible values of the original number (assuming hundredths from 00 to 99). How many are there?

FINAL PROJECT

The precision challenge

★ FINAL PROJECT

The precision challenge

Design a short place-value workshop for a Year 5 visitor. You will teach one idea from this unit so clearly that somebody one year younger can use it.

What you make

- One A4 poster or digital slide with a place-value chart to thousandths.
- Three worked examples: one on place value, one on $\times$ or $\div$ by 10/100/1000, one on rounding.
- A six-question mini test, with a full answer sheet and brief notes on common mistakes.
- One real-world photograph or sketch (market scale, stopwatch, odometer, coin, recipe) with a decimal reading labelled by columns.

How it will be judged

CRITERION	WEIGHT	WHAT EXCELLENCE LOOKS LIKE
Correct mathematics	40%	Every answer and every column label is right
Clear teaching	25%	A Year 5 pupil can follow without you speaking
Real-world link	15%	The labelled reading matches a real instrument
Careful presentation	10%	Columns align; digits are readable
Reflection	10%	You name the mistake you almost made, and how you checked

Work in pairs. Swap mini tests with another pair and mark them. Repair any question that more than half the class miss.



Picture 1.8 Precision tools for a place-value workshop.

CHECK YOUR PROGRESS

Check your progress

Work on your own first. Then compare with a partner.

- 1 Write 506.083 in words, then say the place value of the digit 8.
- 2 Order from smallest to largest: 0.7, 0.07, 0.707, 0.077, 0.7.
- 3 Calculate: a. 3.08×100 b. $42 \div 1000$ c. 0.6×1000
- 4 Decompose 20.405. Then regroup it as thousandths.
- 5 Round 18.5 to the nearest whole number, and 18.45 to the nearest tenth.
- 6 A scale shows 2.465 kg. Write the mass to the nearest tenth of a kilogram and to the nearest kilogram.
- 7 Explain the mistake: " $0.45 \times 10 = 0.450$ because I added a zero."

- 8 Using digits 4, 0, 9, 2 once each, make the largest number less than 50 with exactly two decimal places.

✓ **NOW I CAN**

- ☒ name every column from a million down to thousandths
- ☒ say the place value of any digit in a decimal number
- ☒ multiply and divide by 10, 100 and 1000 by sliding digits
- ☒ compose, decompose and regroup numbers with up to three decimal places
- ☒ round to the nearest whole number and to the nearest tenth
- ☒ check a method a second way, not only by repeating the first way

■ **KEY WORDS**

place value · tenth · hundredth · thousandth · decimal point · compose · decompose · regroup · round · nearest whole number · nearest tenth

UNIT 2 · NUMBER AND SEQUENCE

2

Patterns that keep going

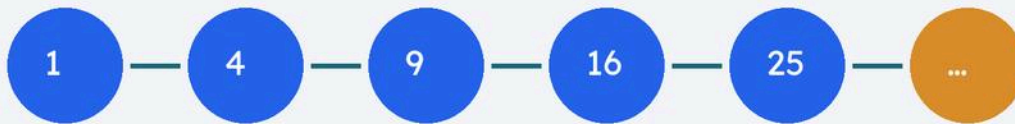
Sequences, squares, cubes, factors and multiples

🕒 **BY THE END YOU WILL**

- ✓ count on and back with fractions and decimals
- ✓ find and use a position-to-term rule
- ✓ work out square and cube numbers and use the 2 notation
- ✓ find common multiples and common factors

Unit 2 · Sequences

Patterns that keep going



Picture 2.0 Square numbers grow like steps: 1, 4, 9, 16, 25...

■ OPEN THE WORKSHOP

A sequence is a list that follows a rule. Once you know the rule, you can find the next term, the tenth term, or a term far down the line without writing every step.

UNIT 2 - TOPIC 2.1

Counting and sequence

Here is how it works

Count on means move forward along a number line. **Count back** means move the other way. The step can be a whole number, a fraction or a decimal.

Examples of constant steps:

- count on in 0.2: 3.4, 3.6, 3.8, 4.0, 4.2

- count back in $\frac{1}{4}$: 2, $1\frac{3}{4}$, $1\frac{1}{2}$, $1\frac{1}{4}$, 1

A **position-to-term rule** tells you the value of the term when you know its position.

If the n^{th} term is $3n + 1$, then:

- position 1 $\rightarrow 3 \times 1 + 1 = 4$
- position 5 $\rightarrow 3 \times 5 + 1 = 16$
- position 10 $\rightarrow 3 \times 10 + 1 = 31$

■ WATCH ME WORK ONE OUT

A sequence starts at 0.5 and counts on in steps of 0.25. What is the 6th term?

- 1 Term 1 = 0.5
- 2 Each new term adds 0.25
- 3 Term 6 = $0.5 + 5 \times 0.25 = 0.5 + 1.25 = \mathbf{1.75}$

⇌ CHECK IT A DIFFERENT WAY

Write the list: 0.5, 0.75, 1.0, 1.25, 1.5, **1.75**. Six terms, same answer.

• YOUR TURN

- 1 Count on four steps from 2.3 in steps of 0.4.
- 2 Count back five steps from 5 in steps of $\frac{1}{2}$.
- 3 The n^{th} term is $2n - 1$. Find terms 1, 4 and 12.
- 4 The n^{th} term is $0.5n + 1$. Find term 8.
- 5 A sequence is 1.1, 1.4, 1.7, 2.0, ... What is the step? What is the next term?
- 6 Inês makes a sequence with position-to-term rule $4n$. Is 30 in her sequence? Explain.

UNIT 2 - TOPIC 2.2

Special numbers

Here is how it works

A **square number** is what you get when you multiply a whole number by itself.

POSITION	CALCULATION	SQUARE NUMBER
1st	1×1	1
2nd	2×2	4
3rd	3×3	9
9th	9×9	81
12th	12×12	144

We write n^2 for "n squared". So $7^2 = 49$.

A **cube number** multiplies a number by itself twice more: $n \times n \times n$, written n^3 .

Cubes up to 5:

- $1^3 = 1$

- $2^3 = 8$
- $3^3 = 27$
- $4^3 = 64$
- $5^3 = 125$

■ WATCH ME WORK ONE OUT

What is the 9th square number?

$9 \times 9 = 81$. Using notation: $9^2 = 81$.

✦ NUMBER FACT

The word *square* is not a metaphor only: 9 dots can make a 3-by-3 square array, and 16 dots make a 4-by-4 array. That is why they are called square numbers.

• YOUR TURN

- 1 Write the first ten square numbers.
- 2 Calculate: a. 6^2 b. 11^2 c. 15^2
- 3 Which of these are square numbers: 20, 25, 36, 40, 49, 50?
- 4 Work out: a. 3^3 b. 4^3 c. 5^3
- 5 True or false: $5^2 \times 5 = 5^3$. Explain.
- 6 Gonalo says the 10th square number is 100. Is he right?

UNIT 2 - TOPIC 2.3

Common multiples and factors

Here is how it works

A **multiple** of 6 is any number in the 6 times table: 6, 12, 18, 24, ...

A **factor** of 12 is a whole number that divides 12 exactly: 1, 2, 3, 4, 6, 12.

A **common multiple** of 4 and 6 is a number that is a multiple of both (12, 24, 36, ...).

The **lowest common multiple (LCM)** of 4 and 6 is 12.

A **common factor** of 12 and 18 is a factor of both (1, 2, 3, 6). The **highest common factor (HCF)** of 12 and 18 is 6.

■ WATCH ME WORK ONE OUT

Find the LCM and HCF of 8 and 12.

Multiples of 8: 8, 16, 24, 32, 40, ... Multiples of 12: 12, 24, 36, ... LCM = **24**

Factors of 8: 1, 2, 4, 8 Factors of 12: 1, 2, 3, 4, 6, 12 Common factors: 1, 2, 4
→ HCF = **4**

• YOUR TURN

- 1 List the first six multiples of 7 and of 9. Circle any common multiples you see.
- 2 Find the LCM of 5 and 8.
- 3 Find all common factors of 20 and 30, then the HCF.
- 4 Two traffic lights flash every 12 s and every 18 s. After how many seconds do they flash together again (from a shared start)?
- 5 Is 1 a common factor of every pair of whole numbers? Explain.
- 6 Find the HCF of 16 and 24, and the LCM of 16 and 24.

✓ NOW I CAN

- ☒ count on and back with fraction and decimal steps
- ☒ use a position-to-term rule
- ☒ calculate square and cube numbers and use 2
- ☒ find common multiples and common factors

■ KEY WORDS

sequence · term · position-to-term rule · square number · cube number · multiple · factor · common multiple · common factor · LCM · HCF

UNIT 3 · 2D SHAPES

3

Flat shapes, sharp eyes

Quadrilaterals, circles and turn symmetry

◎ BY THE END YOU WILL

- ✓ identify, describe, classify and sketch quadrilaterals
- ✓ name the parts of a circle and draw circles accurately
- ✓ identify and describe rotational symmetry

Unit 3 • 2D shapes

Quadrilaterals, circles, turn symmetry



Picture 3.0 Flat shapes you will sort, draw and turn.

■ OPEN THE WORKSHOP

A **2D shape** lives on a flat surface. This unit trains your eye: name it, sort it, draw it, and say how it turns onto itself.

UNIT 3 - TOPIC 3.1

Quadrilaterals

Here is how it works

A **quadrilateral** has four straight sides and four angles.

NAME	WHAT MAKES IT THAT SHAPE
Square	Four equal sides; four right angles
Rectangle	Opposite sides equal; four right angles
Rhombus	Four equal sides; opposite angles equal
Parallelogram	Opposite sides parallel and equal
Trapezium	Exactly one pair of parallel sides (UK)
Kite	Two pairs of adjacent equal sides

Classify means put a shape into the correct group using properties, not only looks.
Sketch means a clear freehand or ruled drawing that shows the properties you claim.

■ WATCH ME WORK ONE OUT

Is every square a rectangle?

A rectangle needs four right angles and opposite sides equal. A square has that, and more (all sides equal). So **yes**: every square is a special rectangle. Not every rectangle is a square.

• YOUR TURN

- 1 Name each property: four equal sides; one pair of parallel sides; four right angles.
- 2 Which quadrilaterals always have four right angles?
- 3 Sketch a parallelogram that is not a rectangle.
- 4 True or false: every rhombus is a square.
- 5 A shape has two pairs of adjacent equal sides and no parallel sides that you can see. Which name fits best?
- 6 Sort these labels into "always", "sometimes", "never" for a parallelogram: right angles; opposite sides equal; four equal sides.

UNIT 3 - TOPIC 3.2

Circles

Here is how it works

PART	MEANING
Centre	The fixed middle point
Radius	Straight line from centre to the edge
Diameter	Straight line across the circle through the centre; twice the radius
Circumference	The distance around the edge
Chord	Straight line joining two points on the circumference
Arc	Part of the circumference

To **draw a circle accurately**, use a pair of compasses: fix the centre, set the radius, and turn steadily.

■ WATCH ME WORK ONE OUT

A circle has radius 3.5 cm. What is the diameter?

Diameter = $2 \times \text{radius} = 2 \times 3.5 = 7 \text{ cm}$.

• YOUR TURN

- 1 Label centre, radius, diameter on a circle you draw.
- 2 Radius 4 cm $\rightarrow$ diameter?
- 3 Diameter 11 cm $\rightarrow$ radius?
- 4 Is a diameter a chord? Is every chord a diameter?
- 5 Draw two circles with the same centre and different radii (concentric).
- 6 Explain why the compass point must not slip when you draw.

UNIT 3 - TOPIC 3.3

Rotational symmetry

Here is how it works

A shape has **rotational symmetry** if it looks the same after a turn of less than a full turn about its centre.

The **order** of rotational symmetry is how many times it matches itself in a full 360° turn.

- A square has order **4** (every 90°).
- An equilateral triangle has order **3** (every 120°).
- A rectangle that is not a square has order **2**.
- A shape with order 1 has **no** rotational symmetry beyond a full turn.

■ WATCH ME WORK ONE OUT

What is the order of rotational symmetry of a regular hexagon?

A regular hexagon matches itself every 60° , six times in a full turn. Order = 6.

• YOUR TURN

- 1 State the order for: square; non-square rectangle; isosceles triangle (not equilateral).
- 2 Sketch a shape with rotational symmetry order 2 that is not a rectangle.
- 3 Does the letter S (in a simple block font) have rotational symmetry? Of what order?
- 4 A windmill sail design matches every 90° . What order is that?
- 5 Can a shape have rotational symmetry order 5? Sketch one idea.
- 6 Explain the difference between line symmetry and rotational symmetry in one sentence each.

✓ NOW I CAN

- ☒ classify and sketch quadrilaterals by properties
- ☒ name circle parts and draw circles with compasses
- ☒ state order of rotational symmetry

■ KEY WORDS

quadrilateral · parallel · radius · diameter · circumference · chord · arc · rotational symmetry · order

UNIT 4 · 3D SHAPES

4

Solids you can hold

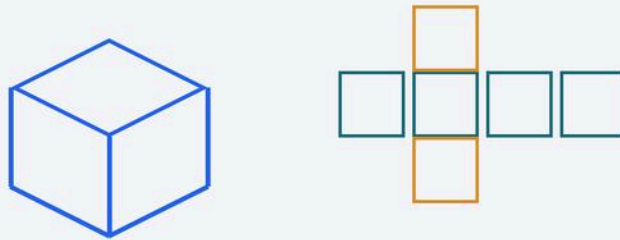
Nets, surface area, capacity and volume

◎ BY THE END YOU WILL

- ✓ identify and sketch compound 3D shapes
- ✓ find different nets for cubes, cuboids, prisms and pyramids
- ✓ link 2D area to surface area of 3D shapes
- ✓ explain the difference between capacity and volume

Unit 4 • 3D shapes

Nets, surface area, capacity and volume



Picture 4.0 Solids, nets and the difference between capacity and volume.

■ OPEN THE WORKSHOP

3D shapes take up space. This unit opens them out into **nets**, counts the skin as **surface area**, and separates how much a container can hold from how much space an object occupies.

UNIT 4 - TOPIC 4.1

Shapes and nets

Here is how it works

A **compound 3D shape** is made by joining simpler solids (for example a cube with a square pyramid on top).

A **net** is a 2D arrangement of faces that folds into the solid without overlaps.

- A cube has 6 square faces. Not every arrangement of six squares folds into a cube.

- A cuboid has 6 rectangular faces (opposite faces equal).
- A prism has two identical end faces and rectangular sides joining them.
- A pyramid has a polygon base and triangular faces meeting at an apex.

Surface area is the total area of all faces. If you paint the outside, surface area is how much paint you need (ignoring thickness).

■ WATCH ME WORK ONE OUT

A cube has edge 4 cm. What is its surface area?

One face: $4 \times 4 = 16 \text{ cm}^2$. Six faces: $6 \times 16 = 96 \text{ cm}^2$.

• YOUR TURN

- 1 Sketch two different nets of a cube.
- 2 How many faces, edges and vertices does a square-based pyramid have?
- 3 A cuboid is 5 cm by 3 cm by 2 cm. Find its surface area.
- 4 Which of these can be a net of a cube? (Draw and test mentally: a straight row of six squares; a cross of five squares with one extra on the side.)
- 5 A compound shape is a cube stuck to a cuboid. List the faces you would **not** paint if they are glued together.
- 6 Explain why surface area uses cm^2 , not cm^3 .

UNIT 4 - TOPIC 4.2

Capacity and volume

Here is how it works

Volume is the amount of space a 3D object occupies. For a cuboid: length $\times$ width $\times$ height. Unit: cm^3 , m^3 .

Capacity is how much a container can hold. Often measured in millilitres or litres (1 litre = 1000 ml). For water, 1 cm^3 has capacity 1 ml.

So a bottle can have **capacity** 500 ml, and the plastic of the bottle itself has a small **volume** of material.

■ **WATCH ME WORK ONE OUT**

A fish tank is 40 cm long, 25 cm wide and 20 cm high (internal). What volume of water fills it to the top?

$40 \times 25 \times 20 = 20\,000 \text{ cm}^3 = \mathbf{20 \text{ litres}}$ (because $1000 \text{ cm}^3 = 1 \text{ litre}$).

● **YOUR TURN**

- 1 Find the volume of a cuboid $6 \text{ cm} \times 4 \text{ cm} \times 3 \text{ cm}$.
- 2 A box holds 2 litres. What is that in millilitres?
- 3 True or false: capacity and volume are always measured in the same unit.
- 4 A mug has capacity 300 ml. Is that the volume of the ceramic? Explain.
- 5 A cube has volume 64 cm^3 . What is the edge length?
- 6 Estimate the capacity of a school water bottle, then check a label if you can.

✓ **NOW I CAN**

- ☒ sketch nets and compound solids
- ☒ calculate simple surface areas
- ☒ distinguish capacity from volume with correct units

■ KEY WORDS

**net · face · edge · vertex · surface area · volume · capacity · prism · pyramid
· cuboid**

UNIT 5 · ADDITION AND SUBTRACTION

5

Directed numbers and letters

Integers, differences and early algebra

◎ BY THE END YOU WILL

- ✓ estimate, add and subtract large integers
- ✓ add and subtract positive and negative integers
- ✓ find the difference between two integers
- ✓ find the value of a letter that represents a number
- ✓ treat an unknown as a variable, not always one fixed answer

Unit 5 • Directed numbers

Integers, differences and letters



Picture 5.0 A number line for positive and negative integers.

■ OPEN THE WORKSHOP

Integers include zero, positive and negative whole numbers. Letters can stand for numbers so one calculation can describe many situations.

UNIT 5 - TOPIC 5.1

Positive and negative integers

Here is how it works

On a number line, positives sit to the right of zero and negatives to the left.

- **Add** a positive: move right.
- **Add** a negative: move left (same as subtracting the positive).
- **Subtract** a positive: move left.
- **Subtract** a negative: move right (minus a minus is plus).

The **difference** between two integers is the positive distance between them on the number line.

Estimate first with large numbers: $4987 + 3012$ is about $5000 + 3000 = 8000$.

■ WATCH ME WORK ONE OUT

Calculate $-3 + 7$ and $5 - (-2)$.

$-3 + 7 = 4$ (from -3 , seven steps right). $5 - (-2) = 5 + 2 = 7$.

Difference between -4 and 3 : from -4 to 3 is 7 steps. Difference = 7.

● YOUR TURN

- 1 Estimate then calculate: $6120 + 2895$.
- 2 Calculate: a. $-5 + 9$ b. $4 + (-6)$ c. $-2 - 5$ d. $8 - (-3)$
- 3 Find the difference between -7 and 2 .
- 4 A temperature rises from -3°C to 5°C . By how many degrees does it rise?
- 5 Order from smallest to largest: $-8, 3, -1, 0, 6, -10$.
- 6 True or false: subtracting a negative always makes a number larger. Give an example.

UNIT 5 - TOPIC 5.2

Letters for numbers

Here is how it works

If n stands for a number, then $n + 4$ means "that number plus four". If $n = 10$, then $n + 4 = 14$.

An **unknown** is a letter whose value we do not yet know. A **variable** can take different values in different situations. The letter is not always one fixed secret number forever: in a formula it can change.

■ WATCH ME WORK ONE OUT

If $p = 6$, find $3p - 2$.

$3p$ means $3 \times p = 18$. $18 - 2 = 16$.

• YOUR TURN

- 1 If $a = 5$, find: a. $a + 9$ b. $2a$ c. $20 - a$
- 2 If $m = 8$ and $n = 3$, find $m + n$ and $m - n$.
- 3 The cost in euros of k tickets at €7 each is $7k$. Find the cost when $k = 4$.
- 4 Can x be 2 in one question and 10 in another? Explain using the word variable.
- 5 Find y if $y + 5 = 12$.
- 6 Write an expression for "4 more than t ".

✓ NOW I CAN

- ☒ calculate with negative integers and differences
- ☒ substitute values for letters
- ☒ talk about unknowns and variables

■ KEY WORDS

integer · negative · positive · difference · estimate · unknown · variable · expression · substitute

UNIT 6 · MULTIPLICATION AND DIVISION

6

Big products, careful shares

Estimate, multiply, divide and test divisibility

🕒 BY THE END YOU WILL

- ✓ estimate before multiplying or dividing
- ✓ multiply whole numbers up to 10 000 by 1-digit or 2-digit numbers
- ✓ divide whole numbers up to 1000 by 1-digit or 2-digit numbers
- ✓ use divisibility tests for 3, 6 and 9

Unit 6 · Multiply and divide

Estimate, calculate, divisibility

$$48 \times 21$$

$$\approx 50 \times 20 = 1000$$

Digit tests

$$3 \cdot 6 \cdot 9$$

Picture 6.0 Estimate first, then multiply or divide carefully.

■ OPEN THE WORKSHOP

Estimation is not a guess for lazy people. It is a safety rail: if your exact answer is nowhere near the estimate, something went wrong.

UNIT 6 - TOPIC 6.1

Multiplication

Here is how it works

- 1 **Estimate** by rounding: $48 \times 21 \approx 50 \times 20 = 1000$.
- 2 Multiply using a written method (column, grid or expanded).
- 3 Compare with the estimate.

■ WATCH ME WORK ONE OUT

Calculate 36×24 .

Estimate: $40 \times 20 = 800$. $36 \times 20 = 720$ $36 \times 4 = 144$ $720 + 144 = 864$ (near 800, sensible).

• YOUR TURN

- 1 Estimate then calculate: a. 27×6 b. 45×18 c. 132×7
- 2 A hall has 28 rows of 35 chairs. How many chairs?
- 3 Multiply 406×9 .
- 4 Which is larger: 25×48 or 30×40 ? Estimate first, then check.
- 5 A crate holds 144 bottles. How many bottles in 15 crates?
- 6 Show two different written methods for 23×17 .

UNIT 6 - TOPIC 6.2

Division

Here is how it works

- 1 **Estimate:** $840 \div 28 \approx 840 \div 30 = 28$.
- 2 Divide with a written method.
- 3 Interpret remainders in context (left over, round up for coaches, and so on).

■ WATCH ME WORK ONE OUT

Calculate $756 \div 12$.

$12 \times 60 = 720$ $756 - 720 = 36$ $12 \times 3 = 36$ Total: 63. So **$756 \div 12 = 63$.**

• **YOUR TURN**

- 1 Estimate then calculate: a. $96 \div 4$ b. $540 \div 15$ c. $728 \div 8$
- 2 Share 250 stickers equally among 20 pupils. How many each? How many left?
- 3 A ribbon is 900 cm long. How many 25 cm pieces?
- 4 Divide 1000 by 16.
- 5 Check 48×17 by dividing the product by 17.
- 6 Explain when you must round a remainder up in a real story.

UNIT 6 - TOPIC 6.3

Tests of divisibility

Here is how it works

DIVISIBLE BY	TEST
3	Sum of digits divisible by 3
9	Sum of digits divisible by 9
6	Divisible by both 2 and 3 (even, and digit sum works for 3)

■ **WATCH ME WORK ONE OUT**

Is 372 divisible by 3, by 6 and by 9?

Digit sum: $3+7+2 = 12$, and 12 is divisible by 3 → **yes for 3**. 372 is even → divisible by 2, and by 3 → **yes for 6**. 12 is not divisible by 9 → **no for 9**.

• YOUR TURN

- 1 Test 234 for divisibility by 3, 6 and 9.
- 2 Test 405 for 3, 6 and 9.
- 3 Write a 3-digit multiple of 9 that is not a multiple of 6.
- 4 Why does the test for 6 need two checks?
- 5 Is 999 divisible by 9? By 3? By 6?
- 6 Create a number divisible by 6 but not by 9.

✓ NOW I CAN

- ☒ estimate and calculate multiplications and divisions in the Year 6 range
- ☒ use digit-sum tests for 3, 6 and 9

■ KEY WORDS

estimate · product · quotient · remainder · divisible · digit sum

UNIT 7 · AVERAGES

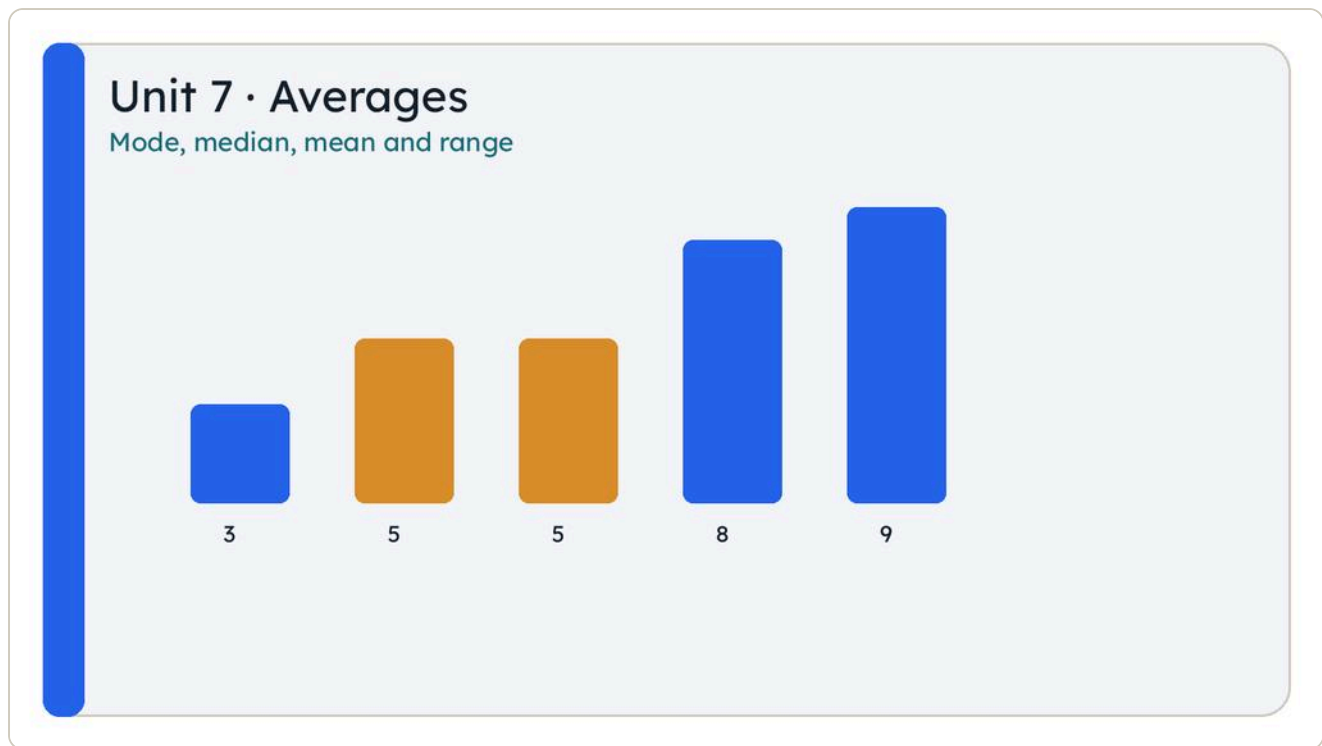
7

One number for a set

Mode, median, mean and range

🕒 BY THE END YOU WILL

- ✓ find the mode and median of a set of data
- ✓ find the mean and range
- ✓ use average and range to describe data and answer questions



Picture 7.0 Bars for a data set: mode, median, mean and range come next.

■ OPEN THE WORKSHOP

An average is a single number that represents a set. Range tells you how spread out the set is. Together they describe both the middle and the width.

UNIT 7 - TOPIC 7.1

Mode, median, mean and range

Here is how it works

For a list of numbers:

- **Mode:** the value that appears most often (there may be more than one mode, or no mode).
- **Median:** the middle value when the list is ordered. For an even count, median is the mean of the two middle values.

- **Mean:** total of all values $\div$ how many values there are.
- **Range:** largest – smallest.

■ WATCH ME WORK ONE OUT

Data: 4, 7, 4, 9, 6

Ordered: 4, 4, 6, 7, 9 Mode = **4** (appears twice) Median = **6** (middle of five)

Mean = $(4+7+4+9+6) \div 5 = 30 \div 5 = 6$ Range = $9 - 4 = 5$

• YOUR TURN

- 1 Find mode, median, mean and range of: 3, 5, 5, 8, 9.
- 2 Scores: 12, 15, 11, 15, 10, 15. Find the mode and the mean.
- 3 Why might the mean not be a whole number even when all data are whole numbers?
- 4 Two classes have the same mean test score but different ranges. What does that tell you?
- 5 Create a set of five numbers with mean 10 and range 6.
- 6 A set has two modes. Give an example and name both modes.

✓ NOW I CAN

- ☒ calculate mode, median, mean and range
- ☒ describe a set using average and spread

■ KEY WORDS

mode · median · mean · range · average · data set · ordered list

UNIT 8 · EXPLORING MEASURES

8

Area and time

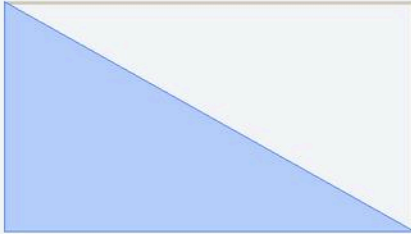
Triangles from rectangles; decimal hours

🕒 **BY THE END YOU WILL**

- ✓ estimate the area of a triangle
- ✓ calculate the area of a triangle using a rectangle
- ✓ convert time intervals between decimals and mixed units

Unit 8 • Measures

Triangle area and time



Area = $\frac{1}{2} \times \text{base} \times \text{height}$

Time: 1.5 h = 1 h 30 min

Picture 8.0 A triangle is half a rectangle when base and height match.

■ OPEN THE WORKSHOP

A right-angled triangle is half of a rectangle. Time can be written as hours and minutes or as a decimal of an hour. This unit moves between those languages.

UNIT 8 - TOPIC 8.1

Rectangles and triangles

Here is how it works

Area of a rectangle = base $\times$ height. A triangle with the same base and height as that rectangle has area **half** of the rectangle:

Area of triangle = (base $\times$ height) $\div$ 2

Estimate first by comparing with a surrounding rectangle.

■ WATCH ME WORK ONE OUT

Triangle base 10 cm, height 6 cm. Find the area.

Rectangle would be $10 \times 6 = 60 \text{ cm}^2$. Triangle = $60 \div 2 = 30 \text{ cm}^2$.

● YOUR TURN

- 1 Area of triangle base 8 cm, height 5 cm.
- 2 Area of triangle base 12 m, height 7 m.
- 3 A triangle has area 24 cm^2 and base 8 cm. What is its height?
- 4 Sketch a triangle inside a rectangle and shade the triangle.
- 5 Estimate the area of a right-angled triangle with legs about 3 cm and 4 cm, then calculate.
- 6 Why must the height be perpendicular to the base?

UNIT 8 - TOPIC 8.2

Time

Here is how it works

1 hour = 60 minutes. $0.5 \text{ h} = 30 \text{ min}$ $0.25 \text{ h} = 15 \text{ min}$ $0.1 \text{ h} = 6 \text{ min}$

To convert **minutes to a decimal hour**: minutes $\div 60$. To convert **decimal hours to minutes**: decimal part $\times 60$.

■ WATCH ME WORK ONE OUT

Write 2.75 hours in hours and minutes.

$$2 \text{ hours} + 0.75 \times 60 \text{ min} = 2 \text{ hours} + 45 \text{ min} = \mathbf{2 \text{ h } 45 \text{ min.}}$$

Write 1 h 18 min as a decimal hour.

$$18 \div 60 = 0.3, \text{ so } \mathbf{1.3 \text{ h.}}$$

● YOUR TURN

- 1 Convert to hours and minutes: a. 1.5 h b. 3.25 h c. 0.2 h
- 2 Convert to decimal hours: a. 2 h 30 min b. 45 min c. 1 h 6 min
- 3 A film lasts 1.75 h. How many minutes is that in total?
- 4 A journey is 90 minutes. Write it as a decimal hour.
- 5 Which is longer: 0.4 h or 25 min?
- 6 A bus timetable shows 0.5 h between stops. How many minutes is that?

✓ NOW I CAN

- ☒ find triangle areas from base and height
- ☒ convert between decimal hours and mixed time units

■ KEY WORDS

area · base · perpendicular height · decimal hour · interval

UNIT 9 · ANGLES

9

Turns you can measure

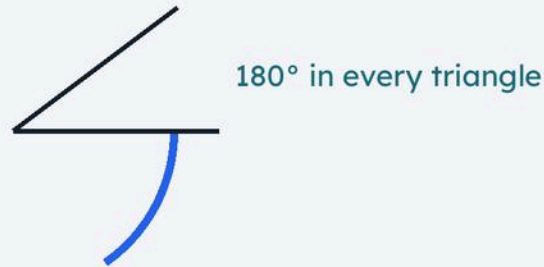
Drawing angles and the triangle sum

🕒 BY THE END YOU WILL

- ✓ measure angles with a protractor
- ✓ draw angles accurately
- ✓ use the sum of angles in a triangle to find missing angles

Unit 9 • Angles

Measure, draw, triangle sum 180°



Picture 9.0 Angles measure turn; every triangle sums to 180° .

■ OPEN THE WORKSHOP

An angle measures turn. A protractor turns that idea into degrees. Inside every triangle, the three angles add to a fixed total.

UNIT 9 - TOPIC 9.1

Measuring and drawing angles

Here is how it works

- Place the protractor centre on the vertex.
- Line the baseline with one arm.
- Read the correct scale (inner or outer) from the other arm.
- Acute: less than 90° . Right: 90° . Obtuse: between 90° and 180° . Straight: 180° .

To **draw** an angle of 65° : draw a baseline, place the protractor, mark 65° , join to the vertex.

■ WATCH ME WORK ONE OUT

Classify 125° .

125 is greater than 90 and less than 180, so it is **obtuse**.

• YOUR TURN

- 1 Classify: 30° , 90° , 150° , 180° .
- 2 Estimate then measure angles in your book margin drawings (teacher may provide).
- 3 Draw angles of 40° , 90° and 135° .
- 4 Two angles on a straight line are 70° and x . Find x .
- 5 Explain one common mistake when reading the wrong protractor scale.
- 6 Draw a reflex angle greater than 180° (if your protractor allows, or construct via full turn).

UNIT 9 - TOPIC 9.2

Angles in a triangle

Here is how it works

The interior angles of any triangle add to **180°** .

If two angles are known, the third is 180° minus their sum.

■ WATCH ME WORK ONE OUT

A triangle has angles 52° and 61° . Find the third angle.

$$52 + 61 = 113 \quad 180 - 113 = 67^\circ$$

• YOUR TURN

- 1 Find the missing angle: 40° and 75° in a triangle.
- 2 Find the missing angle: 90° and 35° .
- 3 Can a triangle have angles 100° , 50° and 40° ? Check the sum.
- 4 An equilateral triangle: what is each angle?
- 5 An isosceles triangle has a vertex angle of 40° . Find the base angles (equal).
- 6 Explain why a triangle cannot have two right angles.

✓ NOW I CAN

- ☒ measure and draw angles
- ☒ find missing triangle angles using 180°

■ KEY WORDS

degree · protractor · acute · obtuse · right angle · interior angle · sum

UNIT 10 · LAWS OF ARITHMETIC

10

Order that matters

Laws, brackets and BIDMAS

🕒 BY THE END YOU WILL

- ✓ use the laws of arithmetic to simplify calculations
- ✓ apply the order of operations
- ✓ use brackets to change the order of operations

Unit 10 · Laws of arithmetic

Order, brackets, BIDMAS

$$3 + 4 \times 5 = 23$$

$$(3 + 4) \times 5 = 35$$

Picture 10.0 Brackets change the answer: order of operations matters.

■ OPEN THE WORKSHOP

Some rearrangements leave a calculation unchanged. Some change it completely. Brackets are the tool that forces the order you mean.

UNIT 10 - TOPIC 10.1

The laws of arithmetic

Here is how it works

Useful laws (with numbers):

- **Commutative** for + and $\times$: $a + b = b + a$; $a \times b = b \times a$
- **Associative** for + and $\times$: $(a + b) + c = a + (b + c)$
- **Distributive**: $a \times (b + c) = a \times b + a \times c$

Order of operations (BIDMAS / BODMAS):

- 1 Brackets
- 2 Indices (powers)
- 3 Division and Multiplication (left to right)
- 4 Addition and Subtraction (left to right)

■ WATCH ME WORK ONE OUT

Calculate $3 + 4 \times 5$ and $(3 + 4) \times 5$.

Without brackets: $4 \times 5 = 20$, then $3 + 20 = 23$. With brackets: $3 + 4 = 7$, then $7 \times 5 = 35$.

⇒ CHECK IT A DIFFERENT WAY

Distribute: $(3 + 4) \times 5 = 3 \times 5 + 4 \times 5 = 15 + 20 = 35$. Same.

• YOUR TURN

- 1 Calculate: a. $10 - 2 \times 3$ b. $(10 - 2) \times 3$ c. $8 \div 2 + 2$
- 2 Insert brackets to make $5 + 3 \times 2 = 16$ true, if possible; if not, say why.
- 3 Simplify $7 \times 13 + 7 \times 7$ using the distributive law.
- 4 True or false: $9 - 4 - 2 = 9 - (4 - 2)$. Check both sides.
- 5 Calculate $2^3 + 4$.
- 6 Rewrite 6×28 as $6 \times (30 - 2)$ and finish.

✓ NOW I CAN

- ☒ use commutative, associative and distributive ideas to simplify
- ☒ apply BIDMAS and brackets correctly

■ KEY WORDS

commutative · associative · distributive · brackets · order of operations · BIDMAS

UNIT 11 • DATA

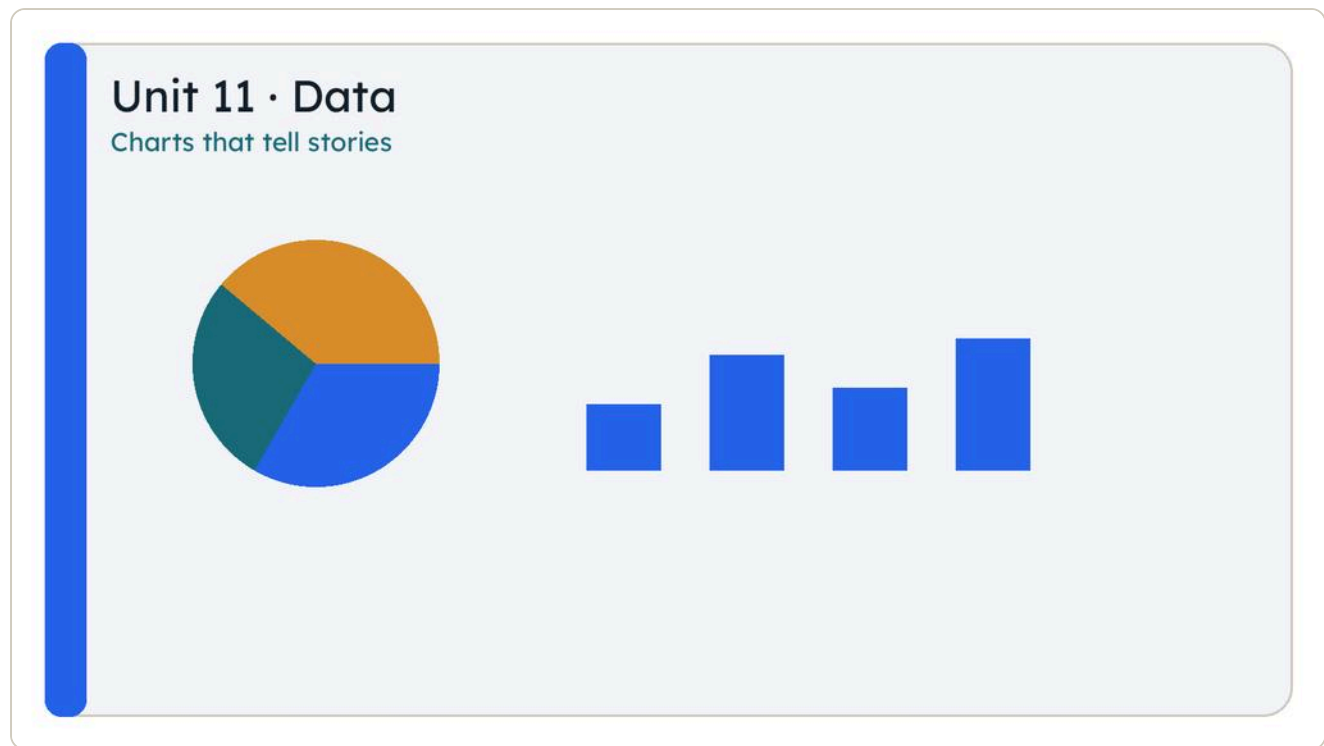
11

Charts that tell stories

From bars and pies to lines and scatter

🕒 **BY THE END YOU WILL**

- ✓ interpret and represent bar charts, dot plots, waffle diagrams and pie charts
- ✓ plan investigations with categories and whole numbers
- ✓ interpret frequency diagrams, line graphs and scatter graphs
- ✓ predict outcomes, look for patterns and check predictions



Picture 11.0 Charts tell different stories about the same investigation.

OPEN THE WORKSHOP

Data is information collected for a reason. A chart is a story about that data. Your job is to read the story and to draw a chart that does not lie.

UNIT 11 - TOPIC 11.1

Bar chart, dot plot, waffle and pie

Here is how it works

CHART	BEST FOR
Bar chart	Comparing categories
Dot plot	Small sets of numerical values
Waffle diagram	Parts of a whole with a grid of cells (often 100)
Pie chart	Parts of a whole as slices of a circle

Investigation steps: ask a question → predict → collect data → represent → check prediction → comment on patterns.

■ WATCH ME WORK ONE OUT

Favourite fruit votes: apple 8, banana 5, orange 7. Total 20. What fraction chose banana?

$5/20 = 1/4$. On a pie chart, banana is a quarter of the circle (90° if full circle is 360°).

• YOUR TURN

- 1 Draw a bar chart for colours: red 4, blue 9, green 5, yellow 2.
- 2 Make a dot plot for scores: 3, 4, 4, 5, 5, 5, 6, 7.
- 3 A waffle of 100 cells shows 30 shaded. What percentage is shaded?
- 4 Plan a one-question investigation for your class (categories only).
- 5 Predict the most popular answer before you collect data.
- 6 Why can a pie chart be misleading if categories are very similar?

UNIT 11 - TOPIC 11.2

Frequency, line and scatter

Here is how it works

- **Frequency diagram:** shows how often values fall into groups.
- **Line graph:** good for data that changes over time (temperature through a day).
- **Scatter graph:** plots pairs (x, y) to look for a relationship (height vs arm span).

Predict → plot → look for trend (upward, downward, none) → check against prediction.

■ WATCH ME WORK ONE OUT

Points (1, 2), (2, 4), (3, 5), (4, 8). Do they suggest an upward trend?

As x increases, y generally increases. **Yes**, a positive trend, though not a perfect line.

• YOUR TURN

- 1 Sketch a line graph of temperature: 08:00 12 °C, 12:00 18 °C, 16:00 17 °C, 20:00 13 °C.
- 2 What story does that graph tell?
- 3 When would a scatter graph be better than a bar chart?
- 4 Collect five pairs of measures (for example handspan and height of classmates) and plot a scatter graph.
- 5 Predict whether your scatter will slope up, down or not at all, then check.
- 6 Explain one pattern you see and one point that does not fit.

✓ NOW I CAN

- ☒ choose and read common charts
- ☒ plan a simple investigation and check a prediction

■ KEY WORDS

bar chart · dot plot · waffle · pie chart · frequency · line graph · scatter graph · trend · prediction

UNIT 12 · FRACTIONS AND PERCENTAGES

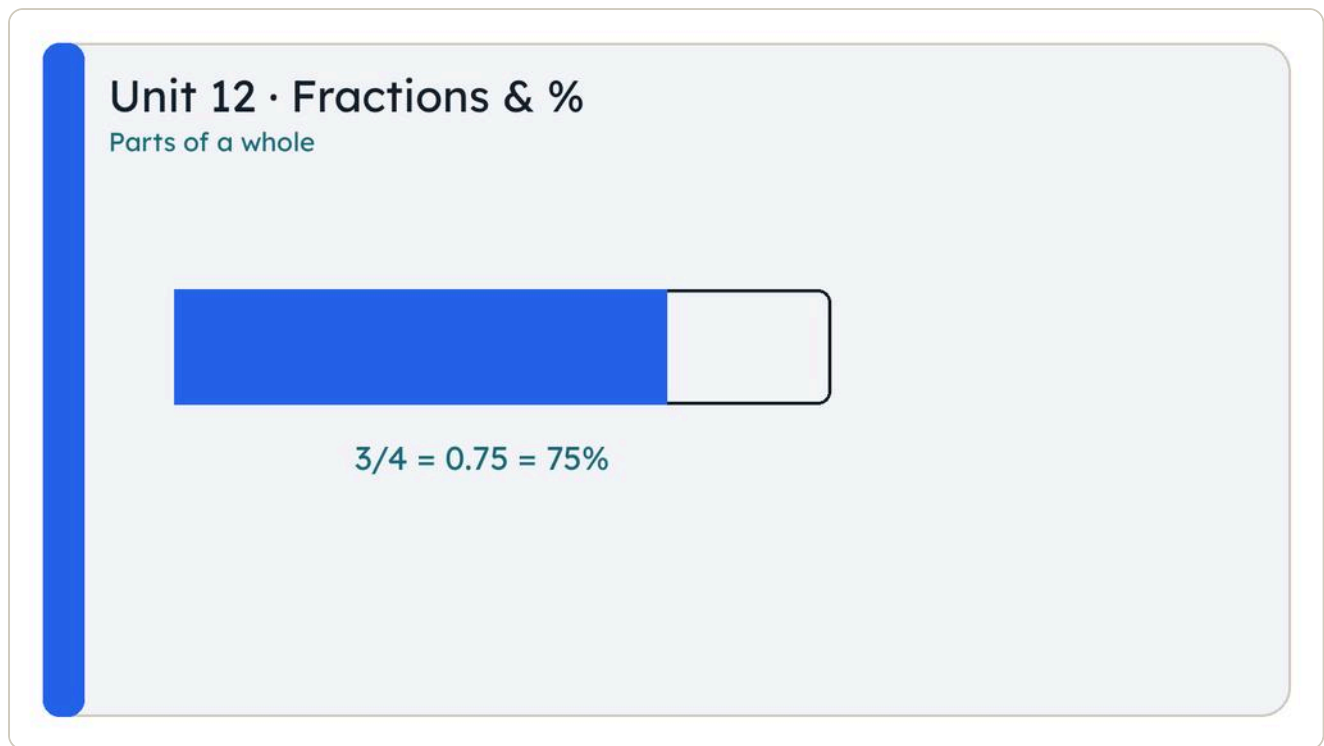
12

Parts of a whole

Fractions, percentages and equivalence

🕒 BY THE END YOU WILL

- ✓ represent proper and improper fractions as division
- ✓ use fractions as operations
- ✓ find percentages of whole numbers and shapes
- ✓ simplify fractions and compare F, D and %



Picture 12.0 Fractions, decimals and percentages name the same idea.

■ OPEN THE WORKSHOP

A fraction is a division in disguise. A percentage is a fraction out of 100. This unit moves fluently between them.

UNIT 12 - TOPIC 12.1

Understanding fractions

Here is how it works

Proper fraction: numerator smaller than denominator ($\frac{3}{4}$). **Improper** fraction: numerator $\geq$ denominator ($\frac{7}{4}$). **Mixed number**: whole + proper part ($1\frac{3}{4}$).

As division: $\frac{3}{4}$ means $3 \div 4 = 0.75$. As an operation: find $\frac{3}{4}$ of 20 $\rightarrow (20 \div 4) \times 3 = 15$.

■ WATCH ME WORK ONE OUT

Write $11/4$ as a mixed number and as a decimal.

$11 \div 4 = 2$ remainder $3 \rightarrow 2 \frac{3}{4}$. 2.75 as a decimal.

● YOUR TURN

- 1 Convert to mixed numbers: $9/2$, $14/5$, $17/3$.
- 2 Convert to improper fractions: $2 \frac{1}{3}$, $4 \frac{2}{5}$.
- 3 Find $2/5$ of 60.
- 4 Find $5/8$ of 32.
- 5 Represent $7/4$ as a division statement.
- 6 Order: $1/2$, $3/4$, $2/3$ (use diagrams or common denominators).

UNIT 12 - TOPIC 12.2

Percentages

Here is how it works

$1\% = 1/100$. $10\% = 1/10$ of a quantity. $50\% = \text{half}$. $25\% = \text{quarter}$. To find 15% of 80:
 $10\% = 8$, $5\% = 4$, total **12**.

■ WATCH ME WORK ONE OUT

Find 35% of 200.

10% of 200 = 20, so $30\% = 60$, $5\% = 10$, total **70**.

• YOUR TURN

- 1 Find 10%, 25% and 50% of 120.
- 2 Find 40% of 90.
- 3 A shape is $\frac{1}{5}$ shaded. What percentage is shaded?
- 4 60% of a class of 30 pupils walk to school. How many walk?
- 5 Which is larger: 30% of 50 or 20% of 80?
- 6 Shade 70% of a 10-by-10 grid.

UNIT 12 - TOPIC 12.3

Equivalence and comparison

Here is how it works

Simplify by dividing numerator and denominator by a common factor: $\frac{12}{18} = \frac{2}{3}$.
Equivalence examples: $\frac{1}{2} = 0.5 = 50\%$; $\frac{1}{4} = 0.25 = 25\%$; $\frac{3}{4} = 0.75 = 75\%$.

Compare using common form (all decimals, all percentages, or common denominators) and the symbols $<$, $>$ or $=$.

■ WATCH ME WORK ONE OUT

Order 0.4, $\frac{3}{5}$ and 45% from smallest to largest.

$0.4 = 40\%$ $\frac{3}{5} = 0.6 = 60\%$ $45\% = 0.45$ Order: **0.4 , 45% , $\frac{3}{5}$**

• YOUR TURN

- 1 Simplify: $8/12$, $15/25$, $9/27$.
- 2 Write as percentages: $1/2$, $1/5$, $3/10$.
- 3 Write as decimals: 25%, 8%, 110%.
- 4 Compare with $<$, $>$ or $=$: $2/5$ $\square$ 0.5; 70% $\square$ $3/4$.
- 5 Order: 0.7, $2/3$, 65%, $3/5$.
- 6 Explain why $50/100$ and $1/2$ are equivalent.

✓ NOW I CAN

- ☒ use fractions as division and as of
- ☒ calculate simple percentages
- ☒ simplify and compare F, D and %

■ KEY WORDS

proper · improper · mixed number · percentage · equivalent · simplify ·
numerator · denominator

UNIT 13 · MULTIPLICATION AND DIVISION 2

13

Fractions and decimals at work

Multiply and divide proper fractions; multiply decimals

◎ BY THE END YOU WILL

- ✓ multiply a proper fraction by a whole number
- ✓ divide a proper fraction by a whole number
- ✓ estimate then multiply decimals (up to 2 d.p.) by 1-digit or 2-digit whole numbers

Unit 13 · Fractions $\times \div$

And decimal multiplication

$$\frac{3}{4} \times 5 = \frac{15}{4}$$

$$2.4 \times 7 = 16.8$$

Picture 13.0 Multiply fractions and decimals with an estimate first.

■ OPEN THE WORKSHOP

Multiplying a fraction by a whole number stacks equal parts. Dividing a fraction by a whole number splits a part into still smaller parts. Decimals follow the same place-value care as Unit 1.

UNIT 13 - TOPIC 13.1

Multiplying and dividing fractions

Here is how it works

Multiply proper fraction by whole number: $\frac{2}{5} \times 3 = \frac{6}{5} = 1 \frac{1}{5}$. Think: three lots of two-fifths.

Divide proper fraction by whole number: $\frac{2}{5} \div 3 = \frac{2}{15}$. Think: share two-fifths into three equal shares, or multiply by $\frac{1}{3}$: $\frac{2}{5} \times \frac{1}{3} = \frac{2}{15}$.

■ WATCH ME WORK ONE OUT

Calculate $\frac{3}{4} \times 5$ and $\frac{3}{4} \div 2$.

$\frac{3}{4} \times 5 = \frac{15}{4} = 3 \frac{3}{4}$. $\frac{3}{4} \div 2 = \frac{3}{8}$.

• YOUR TURN

- 1 Calculate: a. $\frac{1}{5} \times 4$ b. $\frac{2}{3} \times 5$ c. $\frac{3}{8} \times 2$
- 2 Calculate: a. $\frac{1}{2} \div 4$ b. $\frac{3}{5} \div 3$ c. $\frac{5}{6} \div 2$
- 3 A recipe needs $\frac{2}{5}$ kg of flour per batch. How much for 3 batches?
- 4 Share $\frac{3}{4}$ of a pizza equally among 3 people. What fraction each?
- 5 True or false: $\frac{2}{7} \times 3 = \frac{2}{21}$. Correct if false.
- 6 Draw a bar model for $\frac{1}{3} \times 4$.

UNIT 13 - TOPIC 13.2

Multiplying decimal numbers

Here is how it works

- 1 **Estimate** by rounding.
- 2 Multiply as whole numbers, then place the decimal point by counting decimal places, **or** use place-value sliding.

■ WATCH ME WORK ONE OUT

Estimate then calculate 2.4×7 .

Estimate: $2 \times 7 = 14$ (or $2.5 \times 7 = 17.5$). $2.4 \times 7 = \mathbf{16.8}$.

Calculate 1.25×16 .

Estimate: $1 \times 16 = 16$ or $1.3 \times 15 \approx 19.5$. $1.25 \times 16 = 1.25 \times 10 + 1.25 \times 6 = 12.5 + 7.5 = \mathbf{20}$.

● YOUR TURN

- 1 Estimate then calculate: a. 3.5×6 b. 0.8×9 c. 4.25×4
- 2 Calculate 2.6×15 .
- 3 Calculate 0.75×24 .
- 4 A plank is 1.8 m long. What is the total length of 12 such planks?
- 5 Why is 0.5×18 equal to $18 \div 2$?
- 6 Place the decimal point: 36×1.5 should be near 54. Where does the point go in 540 if you forgot it?

✓ NOW I CAN

- ☒ multiply and divide proper fractions by whole numbers
- ☒ multiply decimals by whole numbers with an estimate first

■ KEY WORDS

proper fraction · of · estimate · decimal places · product

UNIT 14 · ADDITION AND SUBTRACTION 2

14

Decimals and fractions together

Regroup, estimate, add and subtract

🕒 BY THE END YOU WILL

- ✓ compose, decompose and regroup decimals to 3 d.p.
- ✓ estimate, add and subtract decimals with same or different lengths
- ✓ add and subtract two fractions with different denominators

Unit 14 · Add & subtract

Decimals and fractions

$$\begin{array}{r} 3.70 \\ + 1.45 \\ \hline \end{array}$$

5.15

$$\frac{1}{3} + \frac{1}{6} = \frac{1}{2}$$

Picture 14.0 Line up points; give fractions a common denominator.

■ OPEN THE WORKSHOP

Line up decimal points. Give fractions a common denominator. Estimation still comes first.

UNIT 14 - TOPIC 14.1

Adding and subtracting decimals

Here is how it works

Compose / decompose / regroup exactly as in Unit 1, then:

- 1 Estimate.
- 2 Write numbers with points aligned.
- 3 Fill empty places with zeros if it helps.
- 4 Add or subtract column by column.

■ WATCH ME WORK ONE OUT

Calculate $3.7 + 1.45$ and $5.2 - 0.86$.

$$3.70 + 1.45 = 5.15 \quad 5.20 - 0.86 = 4.34$$

• YOUR TURN

- 1 Calculate: a. $2.3 + 4.8$ b. $6.05 + 0.7$ c. $9.2 - 3.45$
- 2 Decompose 4.608, then add 0.05.
- 3 Estimate then calculate $12.6 + 7.85$.
- 4 A ribbon is 2.5 m. You cut off 0.75 m. How much is left?
- 5 Add three decimals: $0.6 + 0.35 + 0.08$.
- 6 Explain why 1.4 and 1.40 are equal but may need zeros when subtracting from 3.02.

UNIT 14 - TOPIC 14.2

Adding and subtracting fractions

Here is how it works

Different denominators: find a **common denominator**, convert, then add or subtract numerators.

■ WATCH ME WORK ONE OUT

Calculate $\frac{1}{3} + \frac{1}{6}$ and $\frac{3}{4} - \frac{1}{3}$.

$$\frac{1}{3} + \frac{1}{6} = \frac{2}{6} + \frac{1}{6} = \frac{3}{6} = \frac{1}{2}. \quad \frac{3}{4} - \frac{1}{3}: \text{common denominator } 12 \rightarrow \frac{9}{12} - \frac{4}{12} = \frac{5}{12}.$$

• YOUR TURN

- 1 Calculate: a. $\frac{1}{2} + \frac{1}{4}$ b. $\frac{2}{5} + \frac{1}{10}$ c. $\frac{5}{6} - \frac{1}{3}$
- 2 Calculate $\frac{2}{3} + \frac{1}{4}$.
- 3 Calculate $\frac{7}{8} - \frac{1}{2}$.
- 4 A tank is $\frac{3}{5}$ full. You add $\frac{1}{10}$ more. What fraction is full?
- 5 Which is larger: $\frac{2}{3} - \frac{1}{6}$ or $\frac{1}{2}$?
- 6 Explain why you do not add denominators when adding fractions.

✓ NOW I CAN

- ☒ add and subtract decimals carefully
- ☒ add and subtract fractions with different denominators

■ KEY WORDS

align · common denominator · regroup · estimate · equivalent fraction

UNIT 15 · PROBABILITY

15

How likely?

Proportion, exclusive events and experiments

🕒 BY THE END YOU WILL

- ✓ describe chances using proportion and percentage language
- ✓ recognise mutually exclusive events
- ✓ use likelihood to predict outcomes
- ✓ run simple probability experiments and describe results



Picture 15.0 Likelihood from impossible to certain.

■ OPEN THE WORKSHOP

Probability language sits between impossible and certain. Experiments test whether reality matches the prediction.

UNIT 15 - TOPIC 15.1

Describing and predicting likelihood

Here is how it works

Scale words: impossible → unlikely → even → likely → certain. As proportion: 0 to 1.
As percentage: 0% to 100%.

A fair six-faced die: $P(\text{rolling a 4}) = 1/6$. **Mutually exclusive** events cannot happen at the same time (rolling a 2 and rolling a 5 on one roll).

Experiment: predict → trial many times → record frequencies → compare with prediction.

■ WATCH ME WORK ONE OUT

A bag has 3 red and 5 blue counters. One is drawn at random. What is the probability it is red, as a fraction and a percentage?

Total 8. $P(\text{red}) = 3/8 = 0.375 = 37.5\%$.

• YOUR TURN

- 1 Place on a likelihood scale: sun rising tomorrow; rolling a 7 on a fair six-faced die; drawing a heart from a full shuffled deck (roughly).
- 2 $P(\text{even number on a fair die})$ as a fraction.
- 3 Are "scoring a goal" and "not scoring a goal" in one penalty mutually exclusive? Explain.
- 4 Predict: 20 spins of a fair 2-colour spinner (half red, half blue). About how many red?
- 5 Run 20 coin flips. Record heads. Compare with 50% prediction.
- 6 Why might experimental results differ from the theoretical probability?

✓ NOW I CAN

- ☒ use proportion and percentage chance language
- ☒ identify mutually exclusive events
- ☒ compare experiment with prediction

■ KEY WORDS

probability · likelihood · mutually exclusive · outcome · experiment · theoretical · frequency

UNIT 16 · RATIO AND PROPORTION

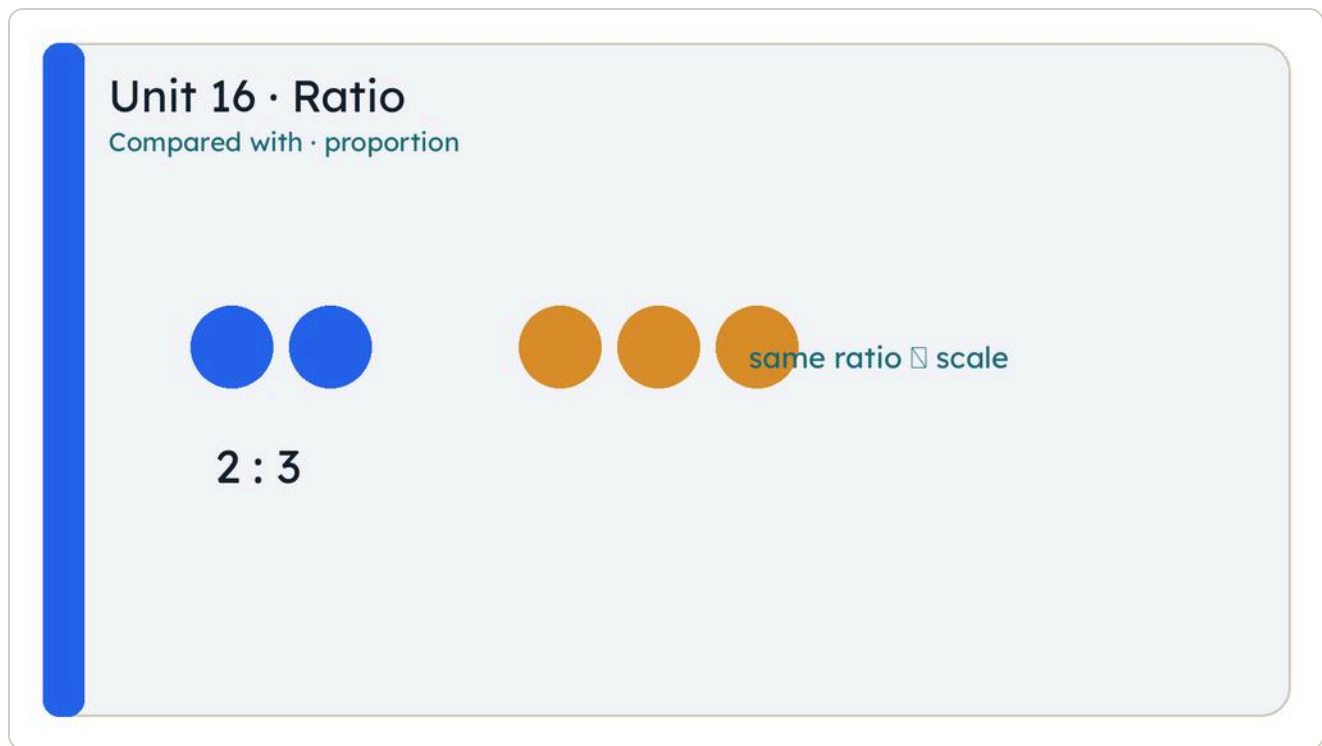
16

Compared with

Equivalent ratios and direct proportion

🕒 **BY THE END YOU WILL**

- ✓ find and use equivalent ratios
- ✓ write a ratio in its simplest form
- ✓ explain what in proportion means
- ✓ scale quantities that change in the same ratio



Picture 16.0 Ratio compares parts; proportion scales them together.

■ OPEN THE WORKSHOP

A ratio compares quantities. Direct proportion means when one multiplies, the other multiplies by the same factor.

UNIT 16 - TOPIC 16.1

Ratio

Here is how it works

The ratio of red to blue paint 2 : 3 means 2 parts red for every 3 parts blue.

Equivalent ratios: $2 : 3 = 4 : 6 = 6 : 9$. **Simplest form:** divide both parts by the HCF.
 $8 : 12 \rightarrow$ divide by 4 $\rightarrow 2 : 3$.

■ WATCH ME WORK ONE OUT

Simplify 12 : 18 and find an equivalent ratio to 2 : 5 with first part 10.

$12 : 18 \div 6 \rightarrow 2 : 3$. $2 : 5 = 10 : 25 (\times 5)$.

• YOUR TURN

- 1 Simplify: a. 4 : 10 b. 9 : 12 c. 15 : 25
- 2 Write three equivalent ratios to 3 : 4.
- 3 A class has 12 girls and 18 boys. Ratio of girls to boys in simplest form?
- 4 Share €40 in the ratio 3 : 5.
- 5 Is 6 : 9 equivalent to 2 : 3?
- 6 A recipe uses flour : sugar = 5 : 2. How much sugar with 250 g flour?

UNIT 16 - TOPIC 16.2

Direct proportion

Here is how it works

If two quantities are **in direct proportion**, multiplying one by k multiplies the other by k .

Example: 3 tickets cost €12. Then 1 ticket costs €4, so 7 tickets cost €28.

■ WATCH ME WORK ONE OUT

4 m of ribbon costs €6. Find the cost of 10 m.

Cost per metre = $6 \div 4 = €1.50$. $10 \text{ m} \rightarrow 10 \times 1.50 = \text{€}15$. (Or scale $\times 2.5$: $4 \times 2.5 = 10$, $6 \times 2.5 = 15$.)

• YOUR TURN

- 1 5 oranges cost €2. How much for 8 oranges (same price each)?
- 2 A car travels 150 km on 10 litres. How far on 4 litres at the same rate?
- 3 If y is proportional to x and $y = 10$ when $x = 2$, find y when $x = 7$.
- 4 Explain why "more workers, less time on a fixed job" is not direct proportion in the same simple way.
- 5 A map scale is 1 : 50 000. What real distance is 2 cm on the map?
- 6 3 identical books cost €18. Cost of 5?

✓ NOW I CAN

- ☒ simplify and scale ratios
- ☒ solve direct proportion problems

■ KEY WORDS

ratio · equivalent · simplest form · parts · direct proportion · scale factor

UNIT 17 · TRANSFORMATION

17

Move, flip, turn

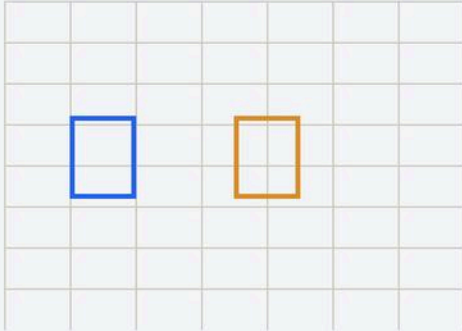
Coordinates, translations, reflections and rotations

© BY THE END YOU WILL

- ✓ read and plot coordinates and form shapes
- ✓ translate shapes on coordinate grids
- ✓ reflect shapes in horizontal, vertical and diagonal mirror lines
- ✓ rotate shapes 90° about a vertex

Unit 17 · Transformation

Move, flip, turn on a grid



translate · reflect · rotate 90°

Picture 17.0 On a grid: translate, reflect and rotate.

■ OPEN THE WORKSHOP

A transformation moves a shape without changing its size in these Year 6 cases. Coordinates name every point so the move is exact.

UNIT 17 - TOPIC 17.1

Coordinates and translations

Here is how it works

A point is written (x, y): x across, y up. To **translate** a shape, slide every vertex by the same vector, for example "3 right, 2 up" or (+3, +2).

■ WATCH ME WORK ONE OUT

Point A(2, 1) is translated 4 right and 3 up. Where is A'?

x: $2 + 4 = 6$ y: $1 + 3 = 4$ A' is (6, 4).

• YOUR TURN

- 1 Plot (0, 0), (2, 0), (2, 3), (0, 3). What shape is it?
- 2 Translate that shape 1 left and 2 down. Give the new vertices.
- 3 Point B(-1, 4) moves to (3, 4). Describe the translation.
- 4 Plot three points that form a right-angled triangle.
- 5 A point is at (5, 2). After translation it is at (5, 7). What happened?
- 6 Explain why every vertex must use the same translation.

UNIT 17 - TOPIC 17.2

Reflections

Here is how it works

A **reflection** flips a shape over a mirror line. Each point and its image are the same perpendicular distance from the mirror, on opposite sides.

Mirrors in this unit: horizontal, vertical, and diagonal (for example $y = x$ style diagonals on a grid).

■ WATCH ME WORK ONE OUT

Point (3, 2) reflected in the y-axis (vertical axis).

x becomes -3, y stays 2 $\rightarrow (-3, 2)$.

• YOUR TURN

- 1 Reflect (4, 1) in the x-axis.
- 2 Reflect (2, 5) in the y-axis.
- 3 Draw a triangle in the first quadrant and reflect it in the y-axis.
- 4 What stays the same under reflection? What changes?
- 5 Reflect (1, 4) in the horizontal line $y = 2$ (count distances).
- 6 Can a reflection change the size of a shape? Explain.

UNIT 17 - TOPIC 17.3

Rotations

Here is how it works

A **rotation** turns a shape about a fixed point (here: a **vertex** of the shape). **90°** is a quarter turn. Direction: clockwise or anticlockwise must be stated.

■ WATCH ME WORK ONE OUT

A square ABCD is rotated 90° clockwise about vertex A. What happens to side AB?

Side AB turns a quarter turn about A to a new position perpendicular to the original AB. Length of AB is unchanged.

• YOUR TURN

- 1 Rotate a point (2, 0) 90° anticlockwise about (0, 0). Where does it land? (Use a grid.)
- 2 Draw an L-triomino and rotate it 90° clockwise about one end.
- 3 How many 90° rotations return a shape to its start?
- 4 Rotate a rectangle 90° about a corner. Is the image still a rectangle?
- 5 Describe a rotation fully (angle, direction, centre).
- 6 True or false: rotation about a vertex always keeps that vertex fixed.

✓ NOW I CAN

- ☒ plot and translate on coordinate grids
- ☒ reflect in horizontal, vertical and diagonal mirrors
- ☒ rotate 90° about a vertex

■ KEY WORDS

coordinates · translate · vector · reflect · mirror line · rotate · clockwise · anticlockwise · vertex

WORD LIST**Word list**

WORD	MEANING	UNIT
place value	the value a digit has because of the column it stands in	1
tenth	one part when a whole is divided into 10 equal parts; 0.1	1
hundredth	one part when a whole is divided into 100 equal parts; 0.01	1
thousandth	one part when a whole is divided into 1000 equal parts; 0.001	1
decimal point	the mark that separates the ones from the tenths	1
compose	put place-value parts together to make a number	1
decompose	split a number into the value of each column	1
regroup	write the same number using a different column unit	1
round	replace a number with a nearby value at a chosen accuracy	1
nearest whole number	the whole number a value is closer to (5 rounds up here)	1
nearest tenth	the tenth a value is closer to (5 hundredths rounds up here)	1

ANSWERS

Answers

Answers are not printed inside the units. Try first, then check here.

Unit 1 • Numbers

1.1 Place value

- 1 a. 70 000 b. 70 c. 0.07 d. 0.007

- 2 a. 403 006 b. 2 000 050 c. 17.08 d. 0.09
- 3 a. three hundred and forty-five thousand, six hundred and seventy-eight
a. one million, two thousand and forty
b. six point three zero five
c. eight hundredths
- 4 3.005, 3.05, 3.15, 3.5, 3.55
- 5 Any number strictly between 4.2 and 4.3, e.g. 4.25
- 6 a. 0.7 (seven tenths > six tenths) b. 12.1 c. 0.1
- 7 0.006 (six thousandths)
- 8 305.209

1.2 $\times$ and $\div$ by 10, 100, 1000

- 1 a. 47; 470; 4700 b. 3.6; 36; 360 c. 520.8; 5208; 52 080 d. 0.09; 0.9; 9
- 2 a. 8; 0.8; 0.08 b. 0.63; 0.063; 0.0063 c. 41.5; 4.15; 0.415 d. 0.901; 0.0901; 0.00901
- 3 a. 100 b. 1000 c. 100 d. 1000
- 4 25 ml
- 5 2500 m
- 6 a. False (not always for decimals) b. True c. True d. False (0.06)
- 7 3.08
- 8 12

1.3 Compose / decompose / regroup

- 1 a. $4+0.6+0.05$ b. $0.3+0.008$ c. $20+7+0.4$ d. $100+30+0.05$ e. $9+0.9+0.09+0.009$ f. 0.07
- 2 a. 68.52 b. 200.46 c. 7.903 d. 1030.7

- 3 a. 46 tenths b. 9 tenths c. 123 tenths d. 50 tenths
- 4 a. 35 hundredths b. 106 hundredths c. 240 hundredths d. 80 hundredths
- 5 a. 125 thousandths b. 400 thousandths c. 1020 thousandths d. 3000 thousandths
- 6 a. 37 b. 62 c. 1500 d. 0.40
- 7 2.50
- 8 d (all of the above)

1.4 Round to nearest whole

- 1 a. 6 b. 7 c. 7 d. 0 e. 1 f. 19 g. 20 h. 100
- 2 1 kg
- 3 2; 3; 3; 4; 4
- 4 a. False (9) b. True c. True d. True
- 5 e.g. 11.5 and 12.4
- 6 No: greater than 7.5 rounds up to 8
- 7 3 km
- 8 €10

1.5 Round to nearest tenth

- 1 a. 4.1 b. 4.2 c. 4.2 d. 0.7 e. 0.8 f. 18.0 g. 18.1 h. 10.0
- 2 12.1 s
- 3 1.2 kg
- 4 a. True b. False (3.2) c. True d. True
- 5 e.g. 5.55 and 5.64
- 6 4.5 → whole 5; 4.55 → tenth 4.6

- 7 0.8 m; 1 m
- 8 After rounding to tenth: 2.4, 2.5, 2.5, 2.4, 2.5 (then order those results as needed)

Unit 2 • Number and sequence

2.1 1. 2.7, 3.1, 3.5, 3.9 2. $4\frac{1}{2}$, 4, $3\frac{1}{2}$, 3, $2\frac{1}{2}$ 3. 1, 7, 23 4. 5 5. step 0.3; next 2.3 6. No (30/4 not whole)

2.2 1. 1,4,9,16,25,36,49,64,81,100 2. 36; 121; 225 3. 25,36,49 4. 27; 64; 125 5. True ($5^2 \times 5 = 5^3$) 6. Yes

2.3 1. common mult e.g. 63 2. 40 3. 1,2,5,10; HCF 10 4. 36 s 5. Yes 6. HCF 8; LCM 48

Unit 3 • 2D shapes

3.1 1. rhombus/square; trapezium; square/rectangle 2. square, rectangle 3. open sketch 4. False 5. kite 6. sometimes; always; sometimes

3.2 1. open labels 2. 8 cm 3. 5.5 cm 4. Yes; no 5. open 6. centre drifts → wrong circle

3.3 1. 4; 2; 1 2. open 3. often order 2 4. 4 5. open 6. line = flip; rotational = turn

Unit 4 • 3D shapes

4.1 1. open nets 2. 5 faces, 8 edges, 5 vertices 3. 62 cm^2 4. row of 6 no; cross often yes 5. glued faces not painted 6. area of surface

4.2 1. 72 cm^3 2. 2000 ml 3. False 4. No (capacity is inside space) 5. 4 cm 6. open estimate

Unit 5 • Addition and subtraction 1

5.1 1. ~9000; exact 9015 2. a. 4 b. -2 c. -7 d. 11 3. 9 4. 8° 5. -10, -8, -1, 0, 3, 6 6. True for integers as taught

5.2 1. 14; 10; 15 2. 11; 5 3. €28 4. Yes (variable) 5. 7 6. $t+4$

Unit 6 • Multiplication and division 1

6.1 1. 162; 810; 924 2. 980 3. 3654 4. $25 \times 48 = 1200$; $30 \times 40 = 1200$ equal 5. 2160 6. open

6.2 1. 24; 36; 91 2. 12 each, 10 left 3. 36 4. 62.5 5. product 816 6. open (e.g. coaches)

6.3 1. 234: yes 3, yes 6, no 9 2. 405: yes 3, no 6, yes 9 3. e.g. 117 4. must be even and $\div 3$ 5. yes 9 and 3; no 6 6. e.g. 12

Unit 7 • Averages

- 1 mode 5; median 5; mean 6; range 6
- 2 mode 15; mean 13
- 3 mean can be not whole when total not divisible
- 4 same centre, different spread
- 5 open (mean 10, $\text{max} - \text{min} = 6$)
- 6 open bimodal

Unit 8 • Measures

8.1 1. 20 cm^2 2. 42 m^2 3. 6 cm 4. open 5. $\sim 6 \text{ cm}^2$; exact 6 6. height perpendicular

8.2 1. 1h30; 3h15; 12 min 2. 2.5; 0.75; 1.1 3. 105 min 4. 1.5 h 5. $0.4 \text{ h} = 24 \text{ min} > 25$? No $25 > 24$ so 25 min longer 6. 30 min

Unit 9 • Angles

9.1 1. acute; right; obtuse; straight 2–3 open 4. 110° 5. open 6. open

9.2 1. 65° 2. 55° 3. No ($100 + 50 + 40 = 190$, not 180)

- 1 60° each 5. base angles 70° each 6. two right angles already total 180°

Unit 10 • Laws of arithmetic

- 1 a. 4 b. 24 c. 6
- 2 $(5 + 3) \times 2 = 16$

3 $7 \times (13 + 7) = 140$

4 False (left 3; right 7)

5 12

6 $6 \times 30 - 6 \times 2 = 168$

Unit 11 • Data

11.1 1–2 open charts 3. 30% 4–6 open

11.2 1–2 open 3. when two measures relate 4–6 open investigation

Unit 12 • Fractions and percentages

12.1 1. $4\frac{1}{2}$; $2\frac{4}{5}$; $5\frac{2}{3}$ 2. $\frac{7}{3}$; $\frac{22}{5}$ 3. 24 4. 20 5. $7 \div 4$ 6. $\frac{1}{2}$, $\frac{2}{3}$, $\frac{3}{4}$

12.2 1. 12; 30; 60 2. 36 3. 20% 4. 18 5. 30% of 50=15; 20% of 80=16 → 16 larger 6. open

12.3 1. $\frac{2}{3}$; $\frac{3}{5}$; $\frac{1}{3}$ 2. 50%; 20%; 30% 3. 0.25; 0.08; 1.1 4. $<$; $<$ 5. $\frac{3}{5}$, 65%, $\frac{2}{3}$, 0.7 6. same value

Unit 13 • Multiplication and division 2

13.1 1. $\frac{4}{5}$; $\frac{10}{3}$; $\frac{6}{8} = \frac{3}{4}$ 2. $\frac{1}{8}$; $\frac{1}{5}$; $\frac{5}{12}$ 3. $\frac{6}{5}$ kg 4. $\frac{1}{4}$ 5. False ($\frac{6}{7}$) 6. open

13.2 1. 21; 7.2; 17 2. 39 3. 18 4. 21.6 m 5. half of 18 6. 54.0

Unit 14 • Addition and subtraction 2

14.1 1. 7.1; 6.75; 5.75 2. 4.658 3. ~20.5; exact 20.45 4. 1.75 m 5. 1.03 6. equal value; zeros hold columns

14.2 1. $\frac{3}{4}$; $\frac{1}{2}$; $\frac{1}{2}$ 2. $\frac{11}{12}$ 3. $\frac{3}{8}$ 4. $\frac{7}{10}$ 5. $\frac{2}{3} - \frac{1}{6} = \frac{1}{2}$ equal to $\frac{1}{2}$ 6. denominators are sizes of parts

Unit 15 • Probability

1 open scale 2. $\frac{1}{2}$ 3. Yes 4. about 10 red 5. open experiment 6. chance variation

Unit 16 • Ratio and proportion

16.1 1. 2:5; 3:4; 3:5 2. e.g. 6:8, 9:12, 12:16 3. 2:3 4. €15 and €25 5. Yes 6. 100 g

16.2 1. €3.20 2. 60 km 3. 35 4. inverse-style story 5. 1 km 6. €30

Unit 17 • Transformation

17.1 1. rectangle 2. $(-1,-2), (1,-2), (1,1), (-1,1)$ 3. 4 right 4. open 5. 5 up 6. same vector

17.2 1. $(4,-1)$ 2. $(-2,5)$ 3. open 4. size same; orientation flips 5. $(1,0)$ 6. No

17.3 1. $(0,2)$ for 90° anticlockwise about origin 2. open 3. 4 4. Yes 5. open 6. True

SOURCES AND REFERENCES

Sources and references

§ SOURCES AND REFERENCES

Curriculum scope

- Prime School, *Y6 Mathematics Syllabus New* (internal scheme of work). Scope and sequence only. All teaching text, examples and exercises in this book are original to Prime Books.

Verified facts

- European Central Bank, common sides of euro coins: €2 thickness 2.20 mm;

1 cent thickness 1.67 mm.

<https://www.ecb.europa.eu/euro/coins/common/html/index.en.html>

- BIPM / SI: since 1983 the metre is the length of the path travelled by light in vacuum during a time interval of $1/299\,792\,458$ of a second. (See also general references on the 1983 CGPM definition of the metre.)

Images

- Cover and interior art originated for Prime Books Year 6 Mathematics (BOOK COVER/, IMAGES/). Logos composited from RESOURCES/Prime School Logo master files.

Publication

- Original Prime Books content for Prime School programmes of study.

FOR TEACHERS

For teachers

§ FOR TEACHERS

What this edition is

Full Year 6 syllabus draft (Units 1 to 17, Terms 1 to 3), 39 scheme topics. PDF/Input was used for **scope only**. All pupil prose is original Prime Books. Third-party student/work books were not copied. Depth can grow toward the 130 to 140 page year budget; this draft is shorter by design until teachers request expansion.

Prior knowledge assumed

- Tenths and hundredths from Year 5
- Reading and writing whole numbers to at least 100 000
- The idea that each column is ten times the column on its right

Teaching notes

- Prefer "digits move, point stays" over "move the point". The second model collapses when answers drop below 1.
- Halfway rounding: this book uses **5 rounds up** consistently. State that aloud if your department uses banker's rounding elsewhere.
- Portuguese contexts (Cascais market, school track, euros, metric) are deliberate. Keep money and measure metric.
- Every calculation in the worked examples was checked in Python before ship.

Apparatus

Ruler, pencil, squared exercise book. Calculator optional for checking, not for the first attempt at place-value slides.

Assessment

- Formative: Your turn sets in each topic
- Summative: Check your progress (8 items)
- Performance: Final project poster and mini test

British English

pupils, programme, colour, centre, metre, litre, practise (verb), practice (noun). No em-dashes.

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